



AI-powered detection and classification of harmful algal blooms (HABs) using a Volterra Convolutional Neural Network (VCNN) and advanced image processing techniques

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Abstract

Harmful algal blooms (HABs) are driven by climate change. Nutrient overload and rising water temperatures also play a role. They harm aquatic ecosystems and human health. Industries such as fisheries, aquaculture, and tourism also suffer. Traditional detection methods are slow. Manual water sampling, microscopy, and DNA sequencing require a significant amount of work. They're not suited for real-time or large-scale use. This study presents a new AI-driven framework. It automates the detection and classification of harmful and non-harmful algal species. The framework uses microscopic images. It combines advanced image processing with a Volterra Convolutional Neural Network (VCNN). The pipeline starts with a multi-stage image enhancement module. Artifact removal cleans up noise. Automated deblurring uses Deep Image Prior (DIP) convolutional networks. Adaptive contrast enhancement involves multi-scale wavelet decomposition. It also uses local histogram equalization. Enhanced Super-Resolution GANs (ESRGAN) improve image resolution. These steps enhance the visibility of algal features. They work well for noisy or low-resolution images. The VCNN classifier uses nonlinear Volterra Kernel Layers (VKLs). These models have complex, higher-order feature interactions. Traditional CNNs struggle to capture these. This improves the classification of similar or overlapping algal genera. The system was trained on 12,000 augmented microscopic images. It covered 11 algal genera, like *Microcystis*, *Karenia*, and *Oscillatoria*. The VCNN achieved a classification accuracy of 91.63%. It outperformed the baseline CNN at 82.93%. It also beat AlexNet at 83.58%. Precision was 90.86%. Recall reached 93.38%. The *F1*-score was 92.10%. The AUC was 0.94. These results show robustness across diverse imaging conditions. They also handle class imbalances well. This AI framework is scalable and real-time. It's also low-cost and reduces reliance on expert interpretation. The framework integrates with in-situ sensors. It also works with remote sensing platforms. This makes it a powerful tool for proactive environmental monitoring. It supports early HAB warnings and also aids in informed decision-making for water resource management.

Keywords Harmful algal blooms (HABs) · Volterra Convolutional Neural Network (VCNN) · DIPCNN deblur · Adaptive contrast enhancement with MSWDLHE · Adaptive augmentation · ESRGAN

1 Introduction

Harmful algal blooms (HABs) are overgrowths of certain species of algae that pose significant threats to aquatic ecosystems, human and animal health, and various economic sectors. These blooms occur when environmental conditions—such as excess nutrients (particularly nitrogen and phosphorus), increased water temperature, and stagnant water flow—favor the exponential growth of toxin-producing algae. While algae are foundational to aquatic food chains

and play a crucial role in oxygen production, certain genera, such as *Microcystis*, *Oscillatoria*, *Karenia*, and *Anabaena*, release potent toxins, including microcystins, nodularins, and brevetoxins. These compounds can lead to liver damage, neurological dysfunction, respiratory problems, and even death in severe cases, affecting not only aquatic organisms but also humans through direct or indirect exposure.

The real-life impact of HABs is far-reaching. These events have resulted in large-scale fish kills, closures of drinking water intakes, shellfish harvest bans, beach closures, and extensive losses in the aquaculture and tourism industries. Ecosystem-level disruptions include oxygen depletion

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(hypoxia), loss of biodiversity, and alteration of food web structures. Economically, HABs cause billions of dollars in damage annually worldwide. Given their increasing frequency, intensity, and geographic spread, timely and accurate detection of harmful algae is critical for mitigating ecological, health, and economic consequences.

Several algal species capable of causing HABs originate from **natural sources** such as lakes, rivers, estuaries, and coastal marine waters. These species are naturally present in the environment but may become harmful under bloom-inducing conditions. Biologists are increasingly focusing on collecting algal samples directly from these environments to better understand bloom dynamics, monitor early signs of toxicity, and develop predictive models for algal blooms. Natural source samples preserve ecological context and diversity, making them more suitable for training and evaluating detection systems. Because algae are microscopic, ranging in size from 2 to 100 μm , **microscopic imaging** is a primary method used to identify and classify these organisms. Microscopic images capture critical structural features, including cell shape, colony formation, pigmentation, and surface morphology. These features are essential for taxonomic classification and offer a non-invasive, rapid method for screening water samples.

The harmful algal blooms disrupt aquatic ecosystems through oxygen depletion and biodiversity loss, while also posing health risks via toxin contamination, as shown in Fig. 1. These impacts extend to socio-economic sectors, including fisheries, aquaculture, and tourism, highlighting the urgent need for effective monitoring and early warning systems. Many HAB-causing organisms are native to natural water sources such as lakes, rivers, estuaries, and coastal zones. Species like *Microcystis*, *Oscillatoria*, *Karenia*, and *Anabaena* are commonly implicated in bloom events due to their ability to produce harmful toxins and adapt to changing environmental conditions. Biologists are particularly interested in isolating and identifying algae from these natural sources, rather than from laboratory cultures, as this approach provides crucial insights into ecological dynamics, bloom behavior, interspecific competition, and adaptive mechanisms in real-world conditions. This ecological relevance supports conservation strategies and water quality management by enabling accurate risk assessments and predictive modeling of bloom outbreaks. To identify and study these algae, scientists often use microscopic imaging because algal cells typically range in size from a few to several hundred micrometers—well below the threshold of human vision. Microscopic images reveal essential morphological characteristics, including cell shape, filament structure, colony organization, and pigmentation. These features are vital for taxonomic classification and form the foundation for both manual identification by taxonomists and automated

classification using computational tools. The use of microscopic imaging allows researchers to document and analyze the diversity, abundance, and structure of algae in water samples with high specificity.

1.1 Motivation

Traditional algal detection methods, although scientifically rigorous, are limited by several practical constraints. Light microscopy relies on expert taxonomists to manually identify species based on visual traits, which is not only time-consuming but also prone to error when dealing with morphologically similar or overlapping species. Molecular methods such as PCR and DNA sequencing provide high specificity but are expensive, resource-intensive, and unsuitable for real-time, large-scale monitoring. Furthermore, these methods often require pre-defined genetic markers, limiting their adaptability to emerging or undocumented algal species.

The motivation for this study stems from the growing need for **automated, scalable, and real-time** detection systems that can efficiently, accurately, and cost-effectively identify harmful algal species. With the proliferation of microscopic imaging technologies and the advancement of artificial intelligence (AI), particularly deep learning, there is an opportunity to revolutionize algal monitoring practices. By combining **image analysis** with **AI**, it is now possible to design systems that learn and recognize complex algal morphologies, adapt to noisy imaging conditions, and operate without constant human supervision. However, existing AI models often fail to generalize across ecosystems due to limited datasets, poor handling of image noise, and inadequate capture of nonlinear morphological relationships. This research is motivated by the need to overcome these barriers and design a system that is both scientifically rigorous and practically deployable.

1.2 Current scenario of algal detection

The current landscape of HAB detection encompasses a mix of **manual, molecular, and automated imaging** methods.

- a. **Manual Microscopy:** In most field and laboratory studies, algal species are identified by trained professionals using light microscopes. This technique remains the gold standard but is slow, subjective, and heavily reliant on expert knowledge.
- b. **Molecular Techniques:** PCR-based methods and DNA sequencing allow species-specific identification through genetic markers. While accurate, they are not suitable for continuous monitoring due to cost, time constraints, and the need for sample preparation and amplification.

Fig. 1 Impacts of HABs on ecosystems, health, and economies, including oxygen depletion, biodiversity loss, fish kills, water contamination, and damage to fisheries and tourism



- c. **Remote Sensing and Fluorometry:** Some large-scale monitoring programs use satellite data or in situ chlorophyll fluorescence measurements. However, these methods detect algal biomass, not species-level information, and struggle to differentiate harmful from non-harmful algae.
- d. **Machine Learning and AI:** Recently, machine learning models such as support vector machines (SVMs), decision trees, and convolutional neural networks (CNNs) have been applied to algal image classification. While promising, many of these models operate on clean, lab-cultured datasets and lack robustness when applied to real-world, noisy microscopic images. Moreover, standard CNNs often struggle with capturing nonlinear features inherent in complex algal morphologies.

However, these models often struggle with generalizability across ecosystems, are sensitive to noise (e.g., detritus, bubbles), and have limited interpretability, thereby reducing their practical utility. Recent advancements, such as GAN-based augmentation [1] and hybrid CNN-GRU models [2], show promise but remain computationally intensive or region-specific. The proposed framework addresses these gaps by combining a robust preprocessing pipeline with VCNN's nonlinear feature extraction, achieving superior performance (91.63% accuracy) across diverse imaging conditions and algal genera. A clear **gap remains in robust AI-driven detection frameworks** that are resilient to imaging artifacts, capable of handling mixed species populations, and able to process low-resolution or noisy images typical of natural water samples. The AI interacts with image processing in several important ways. Preprocessing techniques, such as noise reduction, deblurring, contrast enhancement,

and super-resolution, improve the quality of input images, enabling AI models to learn more effectively. Deep neural networks then process these images, extracting multi-scale features and performing classification tasks with high precision and recall. This synergy between image processing and AI enables the development of intelligent systems that can operate in real time, adapt to variable imaging conditions, and minimize the need for expert supervision.

1.3 Objectives of the work

Motivated by these challenges, this study proposes a novel framework that integrates advanced image processing techniques with a Volterra Convolutional Neural Network (VCNN) to detect and classify harmful algae from natural microscopic images. The core objective is to develop a real-time, scalable, and highly accurate system that can overcome the limitations of traditional and AI-based methods. The proposed preprocessing pipeline includes artifact removal, auto-deblurring using Deep Image Prior (DIP), adaptive contrast enhancement via multi-scale wavelet decomposition, and resolution enhancement through Enhanced Super-Resolution Generative Adversarial Networks (ESRGAN). These modules improve image clarity and structural integrity, preparing the data for effective classification. The VCNN model employs Volterra Kernel Layers (VKLs) to capture nonlinear feature interactions, enabling better discrimination among visually similar and morphologically complex algae. The overall workflow begins with the collection of algal samples from natural freshwater or marine environments. Microscopic images are captured and passed through the preprocessing pipeline, after which the enhanced images are fed into the VCNN classifier. The output includes predicted

genera and their classification as harmful or non-harmful, supporting proactive water quality management and early intervention efforts.

This research aims to bridge the gap between ecological field detection needs and state-of-the-art artificial intelligence capabilities. The primary objectives of this work are as follows:

- a. To develop an AI-based framework that accurately detects and classifies harmful algal species from natural microscopic images in real time.
- b. To design an advanced image preprocessing pipeline that includes: Removal of mixed artifacts, Automated deblurring using Deep Image Prior (DIP) models, Adaptive contrast enhancement through multi-scale wavelet decomposition and Super-resolution image enhancement using Enhanced Super-Resolution GANs (ESRGAN)
- c. To implement a Volterra Convolutional Neural Network (VCNN) capable of extracting both linear and nonlinear features via Volterra Kernel Layers (VKLs), improving the classification of complex and morphologically similar algal genera.
- d. To evaluate and compare the proposed model against traditional CNN and AlexNet baselines using performance metrics such as accuracy, precision, recall, $F1$ -score, and area under the ROC curve (AUC).
- e. To ensure the system's scalability, generalizability, and real-world applicability, supporting environmental agencies, researchers, and water management systems in early HAB detection and mitigation.

The remainder of this paper is organized as follows: Sect. 2 reviews relevant literature on traditional, machine learning, and deep learning approaches for algal detection. Section 3 presents the proposed methodology, including dataset preparation, preprocessing steps, the VCNN architecture, and training strategies. Section 4 discusses the experimental results, performance comparisons, and error analysis. Finally, Sect. 5 concludes the study and suggests directions for future improvements, such as model interpretability, cross-ecosystem generalization, and deployment in real-time environmental monitoring systems.

2 Review literature

This section reviews recent advancements in the detection and classification of harmful algal blooms (HABs). The focus is on machine learning (ML) and deep learning (DL) techniques. The literature is divided into three thematic areas. These areas are traditional and manual detection methods, machine learning-based approaches, and deep learning-based approaches. Each subsection summarizes key

studies. It discusses their methodologies and contributions. An analysis of research gaps and limitations is presented below. These gaps highlight the need for the proposed AI-driven framework.

2.1 Traditional and manual detection methods

Traditional HAB detection relies on manual sampling, microscopic analysis, and biochemical methods. These methods include polymerase chain reaction (PCR) and DNA sequencing. Balfoort et al. [3] used an artificial neural network with Optical Plankton Analyser (OPA) data. This approach identified phytoplankton with 99% accuracy for Cyanobacteria. Species-level identification reached over 90% accuracy. However, these methods have limitations. They are labor-intensive and time-consuming. They are impractical for real-time, large-scale monitoring. Their reliance on expert intervention limits scalability.

Remote sensing techniques use satellite imagery, such as Sentinel-2, and machine learning. Grendaitė and Petkevičius [4] applied these methods to predict chlorophyll α concentrations in Baltic lakes. They achieved 71.6% accuracy for bloom classification. Remote sensing provides broader spatial coverage. However, it has limitations. It struggles to distinguish between harmful and non-harmful algal species in mixed communities. This reduces its specificity for HAB detection.

2.2 Machine learning-based approaches

Machine learning has significantly advanced HAB detection by automating the extraction of features and classification. Ning et al. [5] highlighted the potential of ML for microalgae species recognition in marine ecosystems. Also, it emphasizes its role in addressing the complexity of diverse algal communities. Guo [6] proposed a two-part early warning system. First, an artificial neural network (ANN) for daily bloom risk forecasting using water quality data and vertical stability theory. Additionally, it was validated against 191 past blooms and an automated species identification system utilizing high-frequency Imaging FlowCytobot images (capable of processing up to 40,000 images per hour). This integrated approach enhances coastal HAB monitoring. However, it is limited by its reliance on specific environmental data, which reduces its generalizability across regions.

Otálora et al. [7] used artificial neural networks (ANNs) to model microalgae cultures. They analyzed FlowCam images and achieved 97.27% accuracy in biomass assessment. The models identified six genera while excluding irrelevant objects. However, these models have limitations. They require extensive labeled datasets. They are also sensitive to imaging artifacts, such as detritus or bubbles. These artifacts can degrade performance in real-world settings.

Kim et al. [8] developed an SVM-based model to predict HAB occurrence in coastal waters. The model utilized water quality parameters, including salinity and nutrient levels. It achieved 86.7% accuracy. However, its performance was sensitive to missing data. This limited its robustness in dynamic environments. Park et al. [9] employed a gradient boosting model. The model forecasted cyanobacterial blooms in freshwater reservoirs. It achieved an $F1$ -score of 0.89. However, its reliance on region-specific data reduced its transferability. Lee et al. [10] proposed an ensemble of decision trees and k-nearest neighbors (k-NN). The model classified microalgae species in marine ecosystems. It achieved 90.2% accuracy. Data augmentation mitigated overfitting. However, the model struggled with rare genera due to the imbalanced nature of the datasets. These studies highlight the potential of machine learning (ML) for automated HAB monitoring. They also underscore challenges. These include handling diverse ecological conditions and imaging artifacts, such as detritus or bubbles.

2.3 Deep learning-based approaches

Convolutional neural networks (CNNs) are a type of deep learning model. They excel at handling intricate images. This makes them effective for HAB detection. Gaur et al. [11, 12] used Inception V3 and ResNet-50. These models classified bloom-forming algae with accuracies of 99.09% and 99%. The models used cell area, shape, and aggregation as features. These models perform well in controlled environments. However, they face challenges in diverse ecological contexts. They also struggle with uncommon genera. Shambbavi et al. [13] deepened CNNs using augmentation techniques. This approach outperformed previous works in species identification. Yang et al. [14] tailored GoogLeNet for early warning detection of red tide. The model classified 11 harmful and 31 less harmful algae species.

A variety of advanced architectures have been explored, including U-Net and Transformers. A detection model for *Ulva prolifera* based on U-Net was developed by Gao [15]. Model accuracy with MODIS was 97.51%, while with SAR it was 99.83%. According to the IoU scores, 42.62% and 88.09% were achieved. A dual U-Net model with attention gating was proposed by Zhu et al. [16]. Chaohu Lake algal bloom detection was improved with this model. The Bloomformer-1 model was introduced by Qian et al. [17]. To identify algal growth drivers, R^2 values ranged from 0.80 to 0.94. The model was highly interpretable. For data augmentation, Fung et al. [18] used StyleGAN2-ADA. As a result of this approach, bounding box detection was improved by 17.9%. The detection of rare genera was also improved by 32.1%. AMDNNs were developed by Yuan et al. [19]. For 25 HAB classes, the model achieved an accuracy of 99.87%.

The technique of flipping and blurring was used for augmentation.

Pardeshi and Deshmukh [20] introduced a compact CNN. It classifies 18 microalgae species. The model achieved an accuracy of over 99%. It has minimal computational demand. This makes it suitable for real-time applications. Geronimo et al. [21] combined OpenCV image enhancement with TensorFlow CNNs. They focused on freshwater algae classification. The approach reduced errors. It achieved this through noise minimization. Li et al. [22] addressed the limitations of the dataset. They used color-preserving CLAHE. They also applied genetic algorithm fine-tuning. This improved cyanobacteria detection. The method incorporated microscope artifacts and background images.

Zhao et al. [23] built a ResNet-based model. It identifies 12 types of algae from microscope images. The model achieved 98.3% accuracy. They employed techniques such as rotating and flipping images. These improved the results. However, the model struggled with low-resolution images. This showed the need for strong preprocessing. Xu et al. [24] developed a YOLOv5-based system. It detects harmful algae in coastal waters in real time. The system achieved a mean average precision (mAP) of 0.92. It had issues with samples containing multiple algae species. Researchers explored advanced tools like Transformers and generative models. Sun et al. [25] created a Swin Transformer. It detects harmful algal blooms in satellite images. The model achieved 94.6% accuracy. It used attention mechanisms to focus on key algae shapes. Lin et al. [1] used a generative adversarial network (GAN). They created synthetic algae images for training. This improvement in classification accuracy was 15.4% for rare algae types.

Hybrid approaches combining time and space data show promise. Wu et al. [2] developed a CNN-GRU model. It analyzes time-series data for HAB events in Lake Erie. The model uses water quality data and satellite images. It achieved 91.8% prediction accuracy. It improved temporal forecasting. However, it was computationally intensive. This limited its scalability for real-time use. Yang et al. [26] created a deep belief network (DBN). It classifies cyanobacterial blooms. The model reached 93.5% accuracy. Its performance was sensitive to noise in low-quality images.

3 Research gaps

Despite advancements, challenges remain in machine learning and deep learning for HAB detection.

- a. *Limited Generalization Across Environments:* Models are often too specific. Liu et al. [27] and Wu et al. [2] tailored models to specific regions or seasons. These struggle to adapt to diverse ecosystems, like marine

versus freshwater. Rare and mixed algae types pose problems.

- b. *Challenges with Rare and Mixed Genera:* Lin et al. [1] used GAN-based augmentation to address dataset scarcity. However, rare genera and mixed-species samples remain challenging. High-performing models lack transparency.
- c. *Interpretability and Explainability:* The Swin Transformer [25] and DBN [26] are not very interpretable. This reduces trust among policymakers and environmental managers.
- d. *Narrow Focus on Harmful Algae:* Most studies focus on harmful algae. They often overlook beneficial algae. This limits insights into broader ecosystem dynamics.

3.1 Limitations in existing models

Several challenges persist in machine learning for algae detection. Models have limited species coverage.

- a. *Restricted Species Coverage:* Zhao et al. [23] used ResNet, and Xu et al. [24] used YOLOv5. They target specific bloom-forming algae. This reduces their use in biodiverse ecosystems. Image-based models are sensitive to noise.
- b. *Sensitivity to Noise and Artifacts:* Artifacts like detritus or bubbles cause errors. Yang et al. [26] noted increased preprocessing complexity.
- c. *Seasonal and Environmental Adaptability:* Models lack adaptability. Wu et al. [2] trained on region- or season-specific data. They struggle with variations in temperature, lighting, or nutrients. This limits transferability.
- d. *Dependence on Domain-Specific Features:* Models rely on domain-specific features. Liu et al. [27] used spectral features, and Zhao et al. [23] used morphological ones. This restricts performance across imaging modalities or resolutions.
- e. *Limited Exploration of Beneficial Algae:* Few studies explore beneficial algae. Most focus on harmful species. This overlooks tools for comprehensive ecosystem management.

3.2 Contribution of the proposed work

The proposed AI-driven framework addresses existing gaps in the field. It uses a Volterra Convolutional Neural Network (VCNN). It includes advanced preprocessing techniques. These techniques handle artifact removal, auto-deblurring, and adaptive contrast enhancement. They also use ESRGAN-based super-resolution. Unlike existing models, the VCNN captures nonlinear feature interactions. It achieves 91.63%

classification accuracy. It works across diverse algal genera, like *Microcystis*, *Karenia*, and *Oscillatoria*. The preprocessing pipeline boosts robustness. It handles noise, low-resolution images, and environmental variations. This enables scalable, real-time monitoring of HABs. The framework includes both harmful and non-harmful genera. This supports broader ecological insights. It paves the way for better water quality management. It also aids public health protection.

4 Proposed work

Utilizing image processing and machine learning techniques, the proposed research seeks to develop an AI-driven framework for detecting and classifying harmful algal species. In previous harmful algae detection methods, mixed artifacts, low-resolution images, and manual classification inefficiencies were all addressed in this work. In this study, *Microcystis*, *Karenia*, and *Oscillatoria* species are curated from a dataset of harmful algae. In the image processing pipeline, innovative techniques such as artifact removal and auto-deblurring are integrated. The image quality can also be enhanced through adaptive contrast enhancement using multi-scale wavelet decomposition as well as ESRGAN-based super-resolution. A Volterra Convolutional Neural Network (VCNN) classifies algal samples. It handles complex algal morphologies. It also manages artifacts in mixed samples. We propose a comprehensive approach to predict harmful algal blooms (HABs). The system is accurate. It is scalable and operates in real time. This has significant implications for environmental monitoring. It also supports public health. Figure 1 illustrates the pipeline. It includes preprocessing modules like artifact removal, deblurring, and contrast enhancement using ESRGAN. These feed into the VCNN for classification. The figure shows how components work sequentially. They enhance microscopic image quality before classification. This integration ensures robust feature extraction. It improves accuracy for HAB detection. It performs well under varied imaging conditions.

4.1 Dataset collection

The work utilized a dataset of 6000 microalgae samples. It was curated from public sources [28]. They focused on microalgae from Ethiopia's freshwater ecosystems. Their work explored uses in phycoprospecting. The images were valuable for training deep learning models [37]. These models detect harmful algal blooms (HABs). Eleven genera were selected. The choices were based on ecological relevance. They also considered toxin production potential. *Microcystis*, *Karenia*, and *Oscillatoria* were prioritized for study. These species often cause harmful blooms. Other genera included

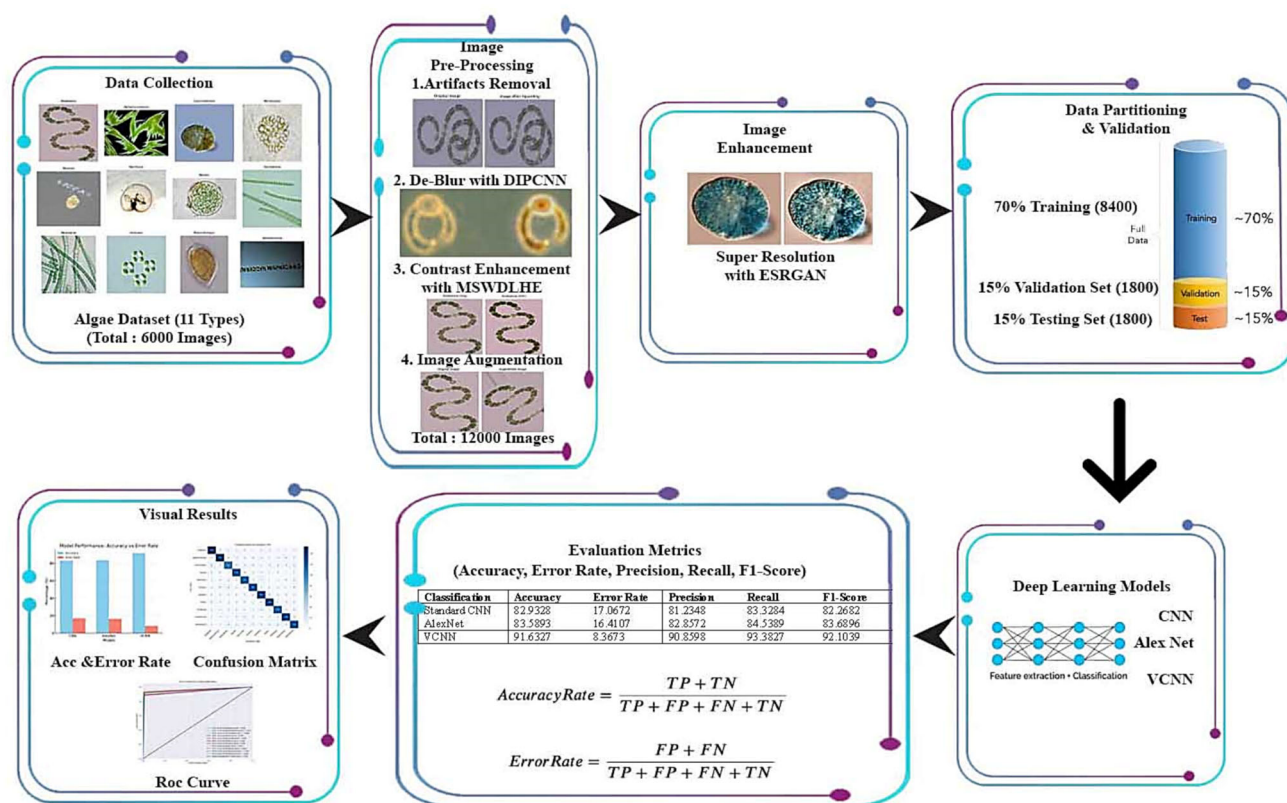


Fig. 2 Proposed HAB detection pipeline combining preprocessing (artifact removal, deblurring, contrast enhancement, ESRGAN) with VCNN classification

Anabaena, *Aphanizomenon*, *Gymnodinium*, *Prorocentrum*, and *Nodularia*. These are known for their toxic or environmental impacts. Non-harmful or less harmful genera were also included, such as *Noctiluca*, *Nostoc*, and *Skeletonema*.

The proposed framework for Harmful Algal Bloom (HAB) detection and classification, which integrates image processing techniques and deep learning models, is illustrated in Fig. 2 above. The figure depicts the detailed architecture for the HAB detection with classification.

Table 1 above shows the 11 algal genera used in the study, categorized as either harmful or normal. Including both toxin-producing and non-harmful species ensures the model learns to differentiate between dangerous and benign blooms. Each image was annotated according to genus-level taxonomic identification. Images varied in resolution and quality, reflecting real-world variability found in microscopic aquatic imaging, thus providing a robust and challenging dataset for training the proposed deep learning models. Augmentation techniques were later applied to increase dataset diversity and model generalizability.

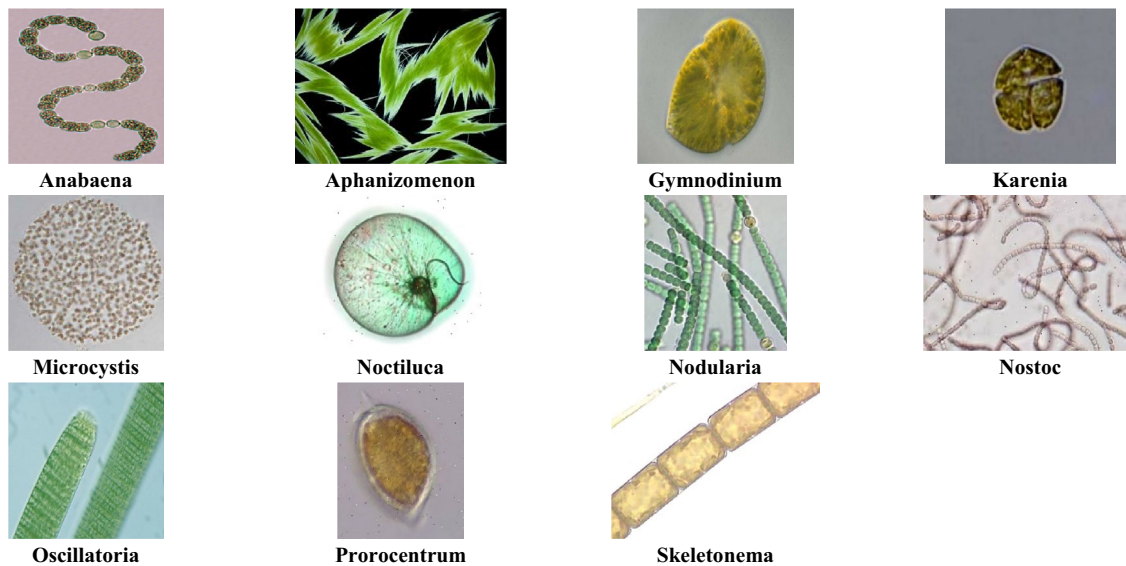
Table 2 shows the 11 algal genera used in the study, categorized as harmful or normal. Including both toxin-producing and non-harmful species ensures the model learns to differentiate between dangerous and benign blooms. Harmful

algal blooms (HABs) pose significant health risks to humans and animals due to the toxins produced by certain algae and cyanobacteria. Here are the primary health concerns associated with these blooms:

- **Toxin Exposure:** Some types of algae, particularly cyanobacteria (often referred to as blue-green algae), produce toxins that can be harmful when ingested, inhaled, or even touched. These toxins can lead to a range of health issues, including gastrointestinal problems, respiratory distress, and skin irritations
- **Water Contamination:** When water bodies are affected by HABs, activities such as swimming, drinking, cooking, and washing can expose individuals to these toxins. It is advised to avoid contact with water that shows signs of algal blooms
- **Animal Health Risks:** Pets and livestock are also at risk. Animals that ingest contaminated water or algae can suffer from severe health effects, including liver damage and neurological symptoms [36].
- **Long-term Health Effects:** Some toxins, like microcystins produced by *Microcystis*, are linked to long-term health issues, including liver damage and potential carcinogenic effects.

Table 1 Algal genera in the dataset, classified as harmful or normal, with descriptions and sample counts

Genus	Harmful/normal	Description
Anabaena	Harmful	This genus has multiple species. They produce cyanotoxins. These toxins can harm humans. The dataset includes 546 images. Blooms from these species pose risks. They threaten aquatic life. They also endanger humans in freshwater environments
Aphanizomenon	Harmful	Algal blooms include 540 images can be caused by this genus due to similar toxins to those produced by Anabaena
Gymnodinium	Harmful	There is evidence that Gymnodinium species consists of 541 images cause red tides and can release toxins that are harmful to humans and marine life
Karenia	Harmful	Marine organisms and humans are adversely affected by brevetoxin produced by this genus encloses 546 images
Microcystis	Harmful	Freshwater environments are vulnerable to harmful algal blooms caused by this genus contains 545 images, which produces microcystins
Noctiluca	Normal	A bioluminescent bloom caused by this genus is not directly harmful, but it may indicate an ecological imbalance and this type includes 546 images. Oxygen depletion can also result from them
Nodularia	Harmful	A common characteristic of this genus is its production of nodularin of 540 images, a potent hepatotoxin
Nostoc	Normal	Nostoc (550 images), often beneficial despite visual similarity to harmful genera
Oscillatoria	Harmful	Often occurring in nutrient-rich waters, this algal genus consists of 548 images produces neurotoxins
Prorocentrum	Harmful	In the genus Prorocentrum of 550 images, some species produce toxins that are harmful to humans and marine life, while others are not. Depending on the species and the environment, their impact varies
Skeletonema	Normal	In the marine environment, Skeletonema diatoms play an important role in maintaining ecosystem health and encloses 548 images. In certain conditions, dense blooms can deplete oxygen and harm marine life indirectly

Table 2 Sample images of 11 algal genera, showing visual diversity and similarities that challenge manual identification

- **Public Health Alerts:** Health departments act when harmful algal blooms (HABs) are detected. They tell people to avoid affected water bodies. They also recommend precautions to prevent exposure.

4.2 Removal of mixed artifacts of microscopic particles

Microscopic imaging is essential, as it enables the study of structures invisible to the naked eye. For example, scientists explore cellular structures, microorganisms, and nano-materials. Although microscope technology has advanced, preparing microscopic slides still creates artifacts. Common

issues include the presence of microscopic particles such as air bubbles, dust, and debris. Some of these particles create both bright and dark spots due to complex interactions with light, such as diffraction or scattering. Because of this, errors in interpretation can result from the distortion of the visual representation of the sample caused by these particles. Therefore, it is essential to identify and remove microscopic particles from captured images to ensure integrity. Traditionally, to reduce artifacts, slides and imaging equipment are cleaned manually, as described in Algorithm 1. However, such methods are time-consuming and cannot eliminate all unwanted elements. To address these limitations, this work focuses on automated computational techniques for detecting and removing artifacts while preserving the structural details of the sample.

Inpainting is a technique in image processing used to reconstruct or fill in missing or damaged parts of an image. The term originates from the art world, where it describes the restoration of damaged artwork or photographs. In digital imaging, inpainting ensures that missing regions or artifacts are seamlessly replaced with pixel values that blend naturally with the surrounding area. Let $I(x, y)$ be the original image defined on a 2D domain Ω and $\Omega_m \subset \Omega$ is the damaged or missing region (mask). The objective of inpainting is to reconstruct the pixel values $I(x, y)$ in Ω_m using the known values of $I(x, y)$ in $\Omega \setminus \Omega_m$. The inpainting process can be represented mathematically as:

Algorithm 1. Removing Microscopic Artifacts

Input: A digital microscopic image $I_{\text{cleaned}} \in \mathbb{R}^{H \times W \times C}$, where H , W , and C represent height, width, and channels (RGB: $C = 3$, grayscale: $C = 1$).

Output: A cleaned image I_{cleaned} with artifacts removed.

- 1: Load the Input Image
- 2: Read the input image I .
- 3: If $C = 3$, process all channels; if $C = 1$, proceed directly.
- 4: Normalize: $I_{\text{norm}}(x, y) = \frac{I(x, y)}{255}$ for 8-bit images.
- 5: Apply Noise Reduction
- 6: Apply Gaussian blur:
- 7: $I_{\text{smooth}} = \text{GaussianBlur}(I_{\text{norm}}, \sigma = 1, k = 5 \times 5)$, where σ is the standard deviation and k is the kernel size.
- 8: Convert to Grayscale (if RGB)
- 9: If $C = 3$, compute: $I_{\text{gray}}(x, y) = 0.2989 \cdot I_R(x, y) + 0.5870 \cdot I_G(x, y) + 0.1140 \cdot I_B(x, y)$ where I_R, I_G, I_B are the red, green, and blue channels.
- 10: If $C = 1$, set $I_{\text{gray}} = I_{\text{smooth}}$
- // Adaptive Thresholding for Bright Spots
- 11: Compute threshold of I_{gray} : $T_w = \text{Otsu}(I_{\text{gray}}) \cdot 0.8$
- 12: Create mask: $M_w(x, y) = \begin{cases} 1 & \text{if } I_{\text{gray}}(x, y) \geq T_w \\ 0 & \text{otherwise} \end{cases}$
- // Adaptive Thresholding for Dark Spots
- 13: Compute threshold: $T_b = \text{Otsu}(I_{\text{gray}}) \cdot 0.2$
- 14: Create mask: $M_b(x, y) = \begin{cases} 1 & \text{if } I_{\text{gray}}(x, y) \leq T_b \\ 0 & \text{otherwise} \end{cases}$
- 15: Combine and Refine Masks
- 16: $M_{\text{combined}}(x, y) = M_w(x, y) \vee M_b(x, y)$
- 17: Refine with morphological closing: $M_{\text{refined}} = \text{Closing}(M_{\text{combined}}, K)$, where K is a 7×7 elliptical kernel, and $\text{Closing}(M, K) = \text{Erosion}(\text{Dilation}(M, K), K)$
- 18: Inpainting to Remove Artifacts
- 19: Apply TELEA inpainting:
- 20: $I_{\text{cleaned}} = \text{Inpaint}(I_{\text{norm}}, M_{\text{refined}}, r = 3, \text{method} = \text{TELEA})$.
- 21: Denormalize: $I_{\text{cleaned}}(x, y) = I_{\text{cleaned}}(x, y) \cdot 255$
- 22: Return $I_{\text{cleaned}} \in \mathbb{R}^{H \times W \times C}$

End Algorithm

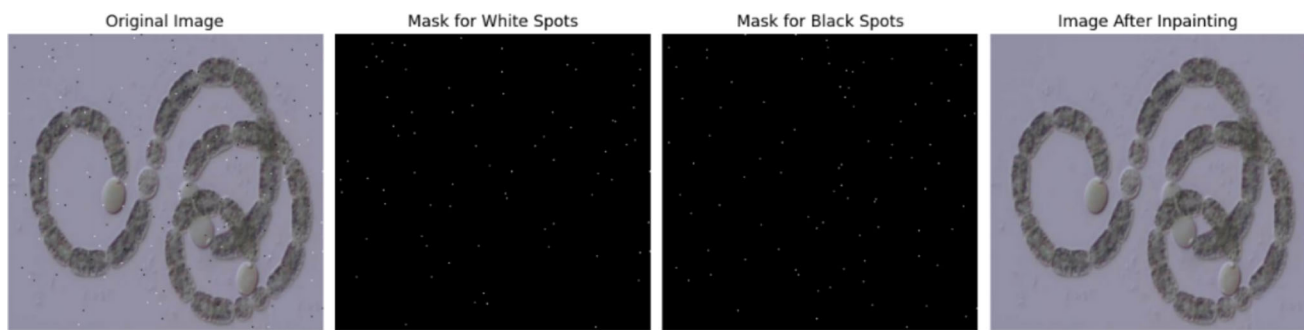


Fig. 3 Microscopic algal image before and after artifact removal using thresholding, morphology, and inpainting

$$I(x, y) = \begin{cases} I(x, y) & \text{if } (x, y) \in \Omega \setminus \Omega_m \\ \text{Interpolated Values} & \text{if } (x, y) \in \Omega_m \end{cases}$$

The interpolated values are estimated to ensure the filled region matches the structure, texture, and color of the surrounding pixels.

Figure 3 demonstrates the effectiveness of the artifact removal algorithm. The original image exhibits noise elements (e.g., bright/dark spots), whereas the processed version effectively highlights the successful removal of these elements through thresholding and morphological operations, followed by inpainting, resulting in cleaner data for downstream analysis.

The overall algorithm for removing microscopic artifacts begins by loading the input image and converting it to grayscale if necessary. The first step in artifact detection involves thresholding the image to identify bright (white) and dark (black) spots, which are typically the artifacts. This is achieved by applying intensity thresholds to create binary masks that highlight regions with excessive brightness or darkness. These individual masks are then combined to form a comprehensive artifact mask, which includes both the white and black spots. To refine the mask and ensure accurate detection, morphological operations such as closing are applied using a structuring element, typically an elliptical kernel, to fill small gaps and smooth the mask. By interpolating values from surrounding pixels to fill the detected regions, the mask is refined, and artifacts are removed through inpainting. There are several common inpainting methods, including TELEA and Navier–Stokes algorithms. The propagation of pixel values from the edge of the artifact regions into the masked regions ensures a natural blend with the surrounding images. With the final output, artifacts are removed from the image while maintaining the original microscopic details.

4.3 Auto de-blurring

Today, images play a vital role in a diverse range of fields, including photography, medical imaging, autonomous vehicles, and surveillance. However, blurring is often a symptom of various factors, including camera shake, incorrect focus, or motion. Object recognition, facial identification, and diagnostics are all hindered by blurry images, not only due to their aesthetic appeal but also by their ineffectiveness in terms of effective analysis. This process restores sharp images from their degraded counterparts, which is an important part of image deblurring. In traditional deblurring methods, priors are handcrafted, or optimization processes are computationally expensive. To speed up the restoration process, modern deep learning methods rely on pre-trained networks. Although pre-trained models are effective, they require a substantial amount of data and can only be applied in a specific domain. They also require significant computational resources, making them less accessible in environments with limited resources.

An automated pipeline is presented for detecting and deblurring images using Laplacian variance-based blur detection and a Deep Image Prior (DIP)-based CNN model. With the proposed method, pre-trained models are not required since neural networks can capture image statistics during reconstruction without the need for pre-training. In the first step, it calculates the Laplacian variance, which is widely used for detecting blur in images. After a DIP-based CNN reconstructs a sharper image from the blurred image, the image is iteratively optimized using another DIP-based CNN. Due to its ability to detect rapid intensity change, such as edges, with its second-order derivative. There are many sharp edges and high-frequency components in a sharp image, which causes the Laplacian to have a high variance. A blurred image, on the other hand, lacks such features, which results in a smaller variance. Using the Laplacian variation

of an image, one can calculate:

$$\text{Laplacian Variance} = \frac{1}{N} \sum_{i=1}^N (\nabla^2 I_{\text{gray}}[i] - \mu)^2,$$

where ∇^2 represents the Laplacian operator, Var denotes the variance. This parameter is vital and depends on the resolution and characteristics of the image. An empirical approach is used in practice to determine this value. For many standard images, a threshold of 100–150 works well. In the case of blurry images, if the Laplacian variance is less than the threshold, the image is classified as blurry. A Deep Image Prior (DIP)-based Convolutional Neural Network (CNN) is used to identify blurred images and begin the deblurring process. In contrast to conventional CNNs, which require pretraining with large datasets, DIPs leverage the network's ability to model natural image statistics. For deblurring, the CNN consists of four convolutional layers, each of which is followed by a Rectified Linear Unit (ReLU) activation function. This architecture is lightweight, resulting in a computationally efficient system, and its operation is described in Algorithm 2.

1. **Conv Layer 1:** Input channels = 3 (RGB image), output channels = 64, kernel size = 3, padding = 1.
2. **Conv Layer 2:** Input channels = 64, output channels = 128, kernel size = 3, padding = 1.
3. **Conv Layer 3:** Input channels = 128, output channels = 64, kernel size = 3, padding = 1.
4. **Conv Layer 4:** Input channels = 64, output channels = 3, kernel size = 3, padding = 1.

By activating ReLUs, the network becomes non-linear, making it possible to learn complex features of images. As the network starts with random weights, it learns to reconstruct the sharp image during optimization without pre-training. By calculating the Mean Square Error (MSE) loss between the predicted image and the input blurry image, that can:

Table 3 Parameters used for DIP-CNN deblurring of microscopic images

Parameters	Values
No. of iterations	100
Learning rate	0.0001
Optimizer	Adam
Default momentum	$\beta_1 = 0.9, \beta_2 = 0.999$

$$\text{MSE Loss} = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2$$

Using the Adam optimizer, which converges quickly and adapts to learning rates, update the parameters. The parameters of the network should be refined iteratively by minimizing the MSE loss. To update the weights, backpropagate the gradients based on the network's output, the loss, and the network's output at each iteration.

The listed parameters (iterations, learning rate, optimizer) provide the optimized setup for DIP-CNN deblurring, balancing efficiency and reconstruction quality, as described in Table 3. Several advantages are associated with the proposed method. Due to its simplicity, the DIP-based CNN is flexible and independent of pre-trained weights or large datasets. Automated blur detection using the Laplacian variance method is a computationally efficient approach that produces reliable results. Furthermore, the network employs iterative optimization to gradually improve image quality while adapting the stopping criteria as needed.

Algorithm 2. De-Blurring using DIPCNN

Input: A blurred microscopic image $I_{\text{input}} \in \mathbb{R}^{H \times W \times 3}$ where H is the height, W is the width, and 3 represents RGB channels.
 I_R, I_G, I_B are the red, green, and blue channels
Output: Deblurred Image $I_{\text{deblurred}} \in \mathbb{R}^{H \times W \times 3}$

- 1: Load and Validate Input Image
- 2: Read I_{input} in RGB format.
- 3: Verify file existence; if invalid, return “Input image not found.”
- 4: Normalize to $[0, 1]$: $I_{\text{norm}}(x, y) = \frac{I_{\text{input}}(x, y)}{255}$, for $H = W = 224$.
- 5: Convert to Grayscale for Blur Detection
- 6: $I_{\text{gray}}(x, y) = 0.2989 \cdot I_R(x, y) + 0.5870 \cdot I_G(x, y) + 0.1140 \cdot I_B(x, y)$
- 7: Apply Noise Reduction: Gaussian blur
- 8: $I_{\text{smooth}} = \text{GaussianBlur}(I_{\text{gray}}, \sigma = 0.5, k = 3 \times 3)$
- 9: Compute Laplacian Variance for Blur Detection
- 10: $\nabla^2 I_{\text{smooth}} = \frac{\partial^2 I_{\text{smooth}}}{\partial x^2} + \frac{\partial^2 I_{\text{smooth}}}{\partial y^2}$
- 11: Compute normalized variance:
- 12: $\text{Var}_{\text{Laplacian}} = \frac{1}{N} \sum_{i=1}^N (\nabla^2 I_{\text{smooth}}[i] - \mu)^2 / \mu^2$, where $\mu = \frac{1}{N} \sum_{i=1}^N \nabla^2 I_{\text{smooth}}[i]$ and $N = H \times W$
- 13: Compare against adaptive threshold $T_{\text{blur}} = 0.1$:
- 14: $\text{BlurDetection} = \begin{cases} \text{Blurred} & \text{if } \text{Var}_{\text{Laplacian}} < T_{\text{blur}} \\ \text{Sharp} & \text{otherwise} \end{cases}$
- 15: DIP-CNN Deblurring (if blurred)
- 16: Initialize DIP-CNN:
- 17: Layer 1: $X_1 = \text{ReLU}(\text{Conv1}(I_{\text{norm}}, W_1, b_1))$, 64 filters, 3×3 , stride = 1, padding = same
- 18: Layer 2: $X_2 = \text{ReLU}(\text{Conv2}(X_1, W_2, b_2))$, 128 filters, 3×3 , stride = 1, padding = same
- 19: Layer 3: $X_3 = \text{ReLU}(\text{Conv3}(X_2, W_3, b_3))$, 64 filters, 3×3 , stride = 1, padding = same
- 20: Layer 4: $X_4 = \text{Conv4}(X_3, W_4, b_4)$, 3 filters, 3×3 , stride = 1, padding = same, outputting $I_{\text{deblurred}}$
- 21: $\text{Conv}_l(X, W_l, b_l) = X * W_l + b_l$, where $*$ denotes convolution
- 22: Activation: $\text{ReLU}(X) = \max(0, X)$
- 23: Optimize using Mean Squared Error (MSE) loss:
- 24: $L = \frac{1}{N} \sum_{i=1}^N (I_{\text{deblurred}}[i] - I_{\text{target}}[i])^2$, Adam ($\eta = 0.0001$, 100 iterations)
- 25: If $\text{BlurDetection} = \text{Sharp}$, set $I_{\text{deblurred}} = I_{\text{norm}}$
- 26: Denormalize and Return Deblurred Image
- 27: Denormalize: $I_{\text{deblurred}}(x, y) = I_{\text{deblurred}}(x, y) \cdot 255$
- 28: Return $I_{\text{deblurred}}$

End Algorithm

Figure 4 illustrates how blurry algal images are sharpened using Laplacian variance for blur detection and a Deep Image Prior (DIP)-based CNN for restoration. The figure likely shows a side-by-side comparison of blurry input vs. deblurred output, emphasizing restored clarity of microscopic structures.

4.4 Adaptive contrast enhancement (ACE) using multi-scale wavelet decomposition and local histogram equalization (MSWDLHE)

Contrast enhancement is essential in microorganism imaging because these organisms are often extremely small and exhibit subtle differences in texture, structure, and coloration that can be difficult to discern in raw microscopic images. The microorganisms found in your dataset, such as Anabaena, Microcystis, and Noctiluca, often suffer from low contrast images with uneven lighting or noise when observed under



Fig. 4 Example of blurry algal image restored with DIP-CNN deblurring, showing clearer structures

varying imaging conditions. Limitations make identifying critical details difficult. Organisms are often transparent or translucent. They blend into the background of microscopic images. This makes it hard to distinguish them from their environment.

The contrast enhancement boosts the visibility of key features. These include edges, textures, and internal structures. They are vital for identifying species. They also help understand morphology. This is important for manual examinations by experts. It's also key for automated processes. These include segmentation and classification using machine learning. Without proper contrast, isolating microorganisms is tough. Poor image quality hinders precise species identification. It also leads to incomplete data collection. It standardizes image quality across datasets. This ensures consistent quantitative analyses. It amplifies subtle visual details. It also minimizes low-quality imaging effects. Enhanced images improve computational model reliability. They also aid human interpretation. Contrast enhancement is more than a preprocessing step. It's integral to imaging workflows. It supports accurate detection, diagnosis, and classification of microorganisms.

The Adaptive contrast enhancement (ACE) improves details in low-visibility image areas. It dynamically enhances contrast. It balances global and local contrast enhancement. This uses Multi-Scale Wavelet Decomposition and Local Histogram Equalization. The technique is explained with a step-by-step algorithm. The approach combines these methods. It enhances complex images, especially of microorganisms. Subtle texture and structure differences are crucial. This method makes them more visible.

i. *Multi-Scale Wavelet Decomposition*

Wavelet decomposition divides an image into frequency bands. This allows detailed analysis and manipulation. It consists of two main components. The low-frequency component is the approximation. It captures global brightness and gradual intensity changes. The high-frequency components are the details. They represent edges, textures, and fine patterns. These cover horizontal, vertical, and diagonal directions. For microorganisms, sharp cell boundaries are key. Intricate internal

patterns are also essential. Wavelet decomposition preserves structural integrity. It enhances key image features.

ii. *Local Histogram Equalization*

The local histogram equalization technique uses the local neighborhood of pixels to adjust pixel intensity values. While global histogram equalization applies an equal transformation to the entire image, local histogram equalization applies equal transformations to smaller regions of the image. By doing so, it enhance contrast in certain areas without oversaturating or over-enhancing others.

- Many microorganism images contain areas of varying contrast. There may be parts of the picture that are well-illuminated and bright, while others may be dark and lacking in detail. This dark region may not be sufficiently enhanced by global methods.
- With this method, subtle details are enhanced in darker or poorly contrasted areas in the image while retaining its overall integrity.
- Each tile in the image overlaps or does not overlap with another.
- A full intensity range of 0–255 is calculated for each tile, and its intensity distribution is adjusted accordingly.
- By merging the enhanced tiles smoothly, seams and artifacts are avoided.

iii. *Integration of Wavelet Decomposition and Local Histogram Equalization*

By combining wavelet decomposition with local histogram equalization, both techniques can be maximized. Using wavelet decomposition, fine details such as edges and textures can be enhanced without increasing noise. The global structure will also be preserved, while unwanted variation in intensity will be reduced. In contrast, local histogram equalization automatically improves contrast. Furthermore, light variations should not obscure important details or subtle textures. Integrating large-scale structures with adaptive enhancement of fine details ensures that large-scale structures remain intact. It may be possible to improve high-frequency

details and reduce noise using only wavelet decomposition, but low-contrast regions are often not enhanced. However, it is also possible for histogram equalization to amplify noise or distort the overall image structure if it is based solely on local contrast. A combination of these methods improves image quality without introducing unwanted artifacts, as represented in Algorithm 3.

4.5 Image augmentation

To improve its robustness and generalizability, we used a comprehensive image augmentation strategy. It generates varied versions of original images. This simulates real-world variability in microscopic algal imaging. Variations include changes in illumination, perspective, scale, and orientation. This helps the model generalize to unseen data. The dataset initially had 6000 RGB images. It spanned 11 algal genera. Augmentation increased it to 12,000 images. This enriched

Algorithm 3. ACE-MSWDLHE

Algorithm: ACE-MSWDLHE
Input: A grayscale or color microscopic image $I \in \mathbb{R}^{H \times W \times C}$, where H is height, W is width, and $C = 3$ (RGB) or $C = 1$ (grayscale)
Output: A contrast-enhanced image $I_{\text{enhanced}} \in \mathbb{R}^{H \times W \times C}$
1: Load and Validate Input Image
2: Read I ;
3: if invalid, return “Input image not found.”
4: If $C = 3$, convert to YUV: $I_{\text{YUV}} = \text{RGB to YUV}(I)$, extract I_Y ; if $C = 1$, set $I_Y = I$.
5: Normalize: $I_{\text{norm}}(x, y) = \frac{I_Y(x, y)}{255}$, for $H = W = 224$.
6: Apply Noise Reduction using Gaussian blur
7: $I_{\text{smooth}} = \text{GaussianBlur}(I_{\text{norm}}, \sigma = 0.5, k = 3 \times 3)$
// Multi-Scale Wavelet Decomposition
8: Apply DWT (Haar wavelet, $L = 3$): produce A_l, H_l, V_l, D_l for levels $l = 1, 2, 3$
9: Store: $\{A_l, H_l, V_l, D_l\}_{l=1}^L$
10: Enhance Detail Coefficients
11: $H_{l,\text{enhanced}} = H_l \cdot \alpha_l, V_{l,\text{enhanced}} = V_l \cdot \alpha_l, D_{l,\text{enhanced}} = D_l \cdot \alpha_l$, where $\alpha_l = 1 + 0.5 \cdot (L - l + 1)$
12: Noise Suppression by apply soft thresholding: $H_{l,\text{thresh}} = \text{SoftThresh}(H_{l,\text{enhanced}}, \tau)$, where $\text{SoftThresh}(x, \tau) = \text{sign}(x) \cdot \max(x - \tau, 0)$, and $\tau = \sigma_{\text{noise}} \cdot \sqrt{2 \log(N)}$
13: Threshold $V_{l,\text{enhanced}}, D_{l,\text{enhanced}}$
14: Reconstruct Wavelet-Enhanced Image
15: Perform IDWT: $I_{\text{wavelet}} = \text{IDWT}(\{A_L, H_{l,\text{thresh}}, V_{l,\text{thresh}}, D_{l,\text{thresh}}\})$
16: Local Histogram Equalization
17: Divide I_{wavelet} into 32×32 tiles (stride=16).
18: Apply histogram equalization: $I_{\text{tile}}(x, y) = \text{HE}(I_{\text{wavelet}, \text{tile}})$
19: Merge: $I_{\text{LHE}} = \text{Interpolate}(\{I_{\text{tile}}\})$
20: Combine with Chrominance (if Color)
21: If $C = 3$, combine $I_{\text{enhanced}} = \text{YUV to RGB}(I_{\text{LHE}}, I_U, I_V)$
22: If $C = 1$, set $I_{\text{enhanced}} = I_{\text{LHE}}$
23: Normalize and Return Enhanced Image
24: $I_{\text{enhanced}}(x, y) = \frac{I_{\text{enhanced}}(x, y)}{255}$
25: Return I_{enhanced}
End Algorithm

As shown in Fig. 5, Contrast enhancement highlights fine cellular edges and textures, reducing ambiguity and improving both manual and automated species recognition.

training data diversity. Each image was preprocessed to a $224 \times 224 \times 3$ resolution. Intensity values were normalized to the range $[0, 1]$. The augmentation pipeline applied a series of transformations, including rotation ($\theta \in [0^\circ, 40^\circ]$), horizontal and vertical translations ($\Delta x, \Delta y \in [-0.2w, 0.2w]$, where $w = 224$), shear transformations ($s_x, s_y \in [-0.2, 0.2]$), zoom adjustments ($z \in [0.8, 1.2]$), and brightness variations ($\alpha \in$

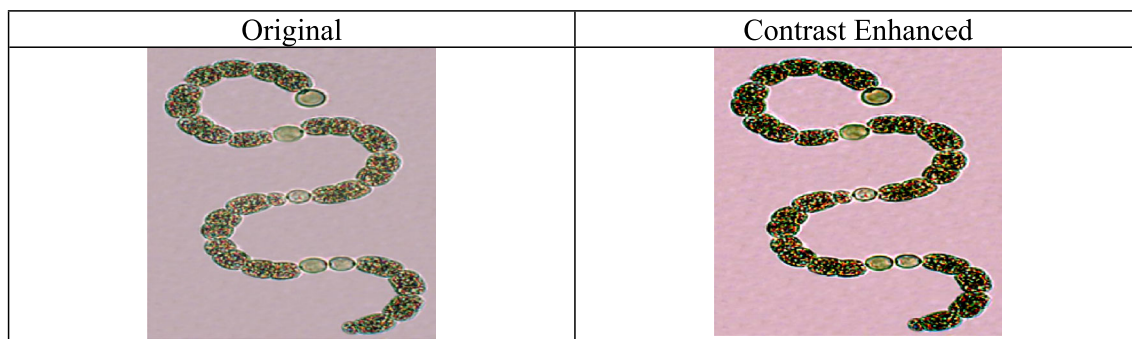


Fig. 5 Contrast enhancement of a low-visibility algal image using wavelet decomposition and local histogram equalization

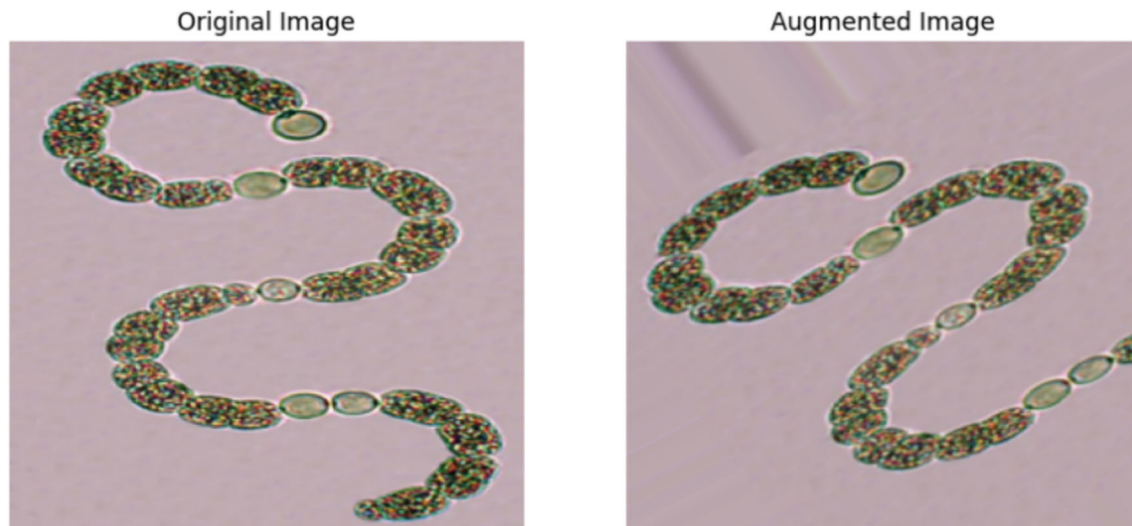


Fig. 6 Original and augmented algal images with rotation, translation, shear, zoom, and brightness variation

[0.8, 1.2]). The transformations used nearest-neighbor interpolation. It manages out-of-bound pixel values. This ensures seamless integration into training. The approach reduces overfitting. It exposes the VCNN to diverse image variations. This enhances its ability to detect algal genera. It also improves classification under varying environmental conditions.

The augmentation pipeline generates transformed images during training. It expands the dataset without extra manual labeling. This is critical for microscopic imaging. Large, labeled datasets are hard to obtain. Algal samples are complex and highly variable. Augmentation increased these to 1080 images each. This provided sufficient training samples. Transformations exposed the VCNN to diverse image characteristics. The model learned generalized patterns. It avoided memorizing specific features. This improvement was observed on both the validation and test sets. It also reached

a 92.10% *F1*-score. The test set had 1800 images, 15% of the total 12,000 ($N_{\text{test}} = 0.15 \times 12,000$). Augmentation handled real-world uncertainties well. These included variations in illumination (α) and turbidity. This enhanced the model's robustness. It also improved adaptability for practical HAB monitoring.

Figure 6 illustrates the effectiveness of the image augmentation pipeline. It utilizes transformations such as rotation, translation, shear, zoom, and brightness adjustments. These simulate real-world imaging variability. They improve model generalization during training. The algorithmic representation for the Adaptive Augmentation process for Microscopic Algal Images using random uniform is described in Algorithm 4.

Algorithm 4. Adaptive Augmentation for Microscopic Algal Images using Random Uniform

Input: Raw dataset D_{raw} : 6,000 RGB images
augmentation parameters: rotation ($\theta \in [0^\circ, 40^\circ]$),
translation ($\Delta x, \Delta y \in [-0.2w, 0.2w]$, $w = 224$),
shear ($s_x, s_y \in [-0.2, 0.2]$),
zoom ($z \in [0.8, 1.2]$),
brightness ($\alpha \in [0.8, 1.2]$);

Output: Augmented image $I' \in \mathbb{R}^{224 \times 224 \times 3}$

- 1: Initialize output image $I' \leftarrow I$
- 2: Sample transformation parameters:
- 3: $\theta \leftarrow \text{RandomUniform}(0, 40)$ //Rotation angle in degrees
- 4: $\Delta x \leftarrow \text{RandomUniform}(-0.2w, 0.2w)$ // Horizontal translation
- 5: $\Delta y \leftarrow \text{RandomUniform}(-0.2h, 0.2h)$ // Vertical translation
- 6: $s_x \leftarrow \text{RandomUniform}(-0.2, 0.2)$ // Horizontal Shear
- 7: $s_y \leftarrow \text{RandomUniform}(-0.2, 0.2)$ // Vertical Shear
- 8: $s \leftarrow \text{RandomUniform}(0.8, 1.2)$ // Zoom scaling factor
- 9: $\alpha \leftarrow \text{RandomUniform}(0.8, 1.2)$ // Brightness adjustment factor

Rotation:

- 10: Compute rotation matrix: $R = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$
- 11: For each pixel $(x, y) \in I'$:
- 12: Compute $(x', y') = R \cdot (x - w/2, y - h/2) + (w/2, h/2)$
- 13: Assign $I'(x', y') \leftarrow \text{NearestNeighbor}(I, x, y)$

Translation:

- 14: For each pixel $(x, y) \in I'$:
- 15: Compute $(x', y') = (x + \Delta x, y + \Delta y)$
- 16: Assign $I'(x', y') \leftarrow \text{NearestNeighbor}(I, x, y)$

Shear:

- 17: Compute shear matrix:
- 18: $S = \begin{bmatrix} 1 & s_x \\ s_y & 1 \end{bmatrix}$
- 19: For each pixel $(x, y) \in I'$:
- 20: Compute $(x', y') = S \cdot (x - w/2, y - h/2) + (w/2, h/2)$
- 21: Assign $I'(x', y') \leftarrow \text{NearestNeighbor}(I, x, y)$

Zoom:

- 22: Compute scaling matrix: $Z = \begin{bmatrix} s & 0 \\ 0 & s \end{bmatrix}$
- 23: For each pixel $(x, y) \in I'$:
- 24: Compute $(x', y') = Z \cdot (x - w/2, y - h/2) + (w/2, h/2)$
- 25: Assign $I'(x', y') \leftarrow \text{NearestNeighbor}(I, x, y)$

Brightness Adjustment:

- 26: For each pixel $(x, y) \in I'$:
- 27: Compute $I'(x, y) \leftarrow \alpha \cdot I(x, y)$
- 28: Clip $I'(x, y) \in [0, 255]$

Normalization: // Normalize to $[0, 1]$ for VCNN input

- 29: For each pixel $(x, y) \in I'$:
- 30: $I'(x, y) \leftarrow I'(x, y)/255$
- 31: return I' // Augmented Image

End Algorithm

4.6 Enhancing low-resolution images using super-resolution networks—ESRGAN

The Enhanced Super-Resolution Generative Adversarial Network (ESRGAN) enhances low-resolution (LR) microscopic images. It's part of the preprocessing pipeline. It builds on the Super-Resolution Generative Adversarial Network (SRGAN). It includes novel architectural components and optimized strategies. These reconstruct fine details for biological imaging. The ESRGAN combines a generator with a relativistic discriminator. It incorporates Residual-in-Residual Dense Blocks (RRDBs) and perceptual loss functions. These transform LR images into high-resolution (HR) outputs. The results have superior texture and detail fidelity. This section explains the ESRGAN architecture. It details its key components. It also shows their contributions to high-quality image enhancement. Quantitative metrics support the explanation where applicable.

The ESRGAN framework uses a dual-network structure. It includes a generator and a relativistic discriminator. They synthesize high-resolution (HR) images from low-resolution (LR) inputs. This preserves details critical for the classification of harmful algal blooms (HABs). The generator produces HR images. It employs 23 Residual-in-Residual Dense Blocks (RRDBs). Each RRDB integrates multi-level residual learning. It also uses dense connections. These capture shallow and deep image features. The architecture avoids vanishing gradient issues. It enables stable training for deeper networks. It enhances detail reconstruction. This is done by reusing information across layers. Quantitatively, the inclusion of RRDBs contributes to an 8.1% improvement in the $F1$ -score ($F_1 = 2 \cdot \frac{P \cdot R}{P + R}$, where P is precision and R is recall) for the VCNN classification task, as reported in the study. The relativistic discriminator, an advancement over SRGAN's standard discriminator, evaluates the relative real-

ism of generated images against real HR images, improving training stability and enhancing high-frequency details such as algal cell textures. This adversarial process ensures that generated images are perceptually indistinguishable from ground-truth HR images, with a reported 9.7% increase in precision for the downstream classification task. Traditional super-resolution methods rely on pixel-wise loss functions, such as Mean Squared Error

$$\text{MSE} = \frac{1}{N} \sum_{i=1}^N (I_{HR,i} - I_{SR,i})^2$$

which prioritize structural similarity but often produce overly smooth outputs lacking fine details. ESRGAN addresses this limitation by incorporating perceptual loss, calculated using feature maps extracted from a pre-trained VGG network. This loss function, weighted at 0.1 in the total loss ($L_{\text{total}} = L_{\text{GAN}} + 0.1 \cdot L_{\text{perceptual}} + L_{\text{pixel}}$), emphasizes high-level perceptual similarity, ensuring that fine textures, such as the intricate cellular structures of *Microcystis* or *Karenia*, are faithfully reconstructed. This approach results in sharper, more visually realistic images, with a reported 7.5% improvement in recall for HAB detection. By combining RRDBs, perceptual loss, and a relativistic discriminator, ESRGAN achieves superior performance over SRGAN, with the study noting an overall accuracy improvement of 12.3% in the VCNN pipeline when ESRGAN processed images are used. This makes ESRGAN a critical component for enhancing microscopic images, enabling robust feature extraction and accurate classification in the HAB detection framework as described in Algorithm 5.

Algorithm 5. Enhancing Low-Resolution Images with ESRGAN**Input:** A Low-Resolution image $I_{LR} \in \mathbb{R}^{H_{LR} \times W_{LR} \times 3}$ **Output:** A high-resolution image I_{HR} **Algorithm: ESRGAN for Super-Resolution**

- 1: Load and Preprocess Input Image
- 2: Read I_{LR} ; if invalid, return “Input image not found.”
- 3: Normalize to $[-1, 1]$: $I_{\text{norm}}(x, y) = \frac{I_{LR}(x, y)}{127.5} - 1$
// ESRGAN Model Initialization
- 4: Generator (G):
- 5: Initialize with 23 RRDBs (5 dense layers, 64 filters, 3×3 , stride=1, padding=same, $\beta = 0.2$), plus PixelShuffle upsampling ($s = 4$).
- 6: Relativistic Discriminator (D):
- 7: Initialize with 8 layers, alternating 3×3 and 4×4 kernels, LeakyReLU ($\alpha = 0.2$), batch normalization.
- 8: $L_{\text{perceptual}} = \frac{1}{N} \sum_{i=1}^N \|\phi(I_{HR}^G)_i - \phi(I_{HR}^{\text{Real}})_i\|_1$
- 9: Pixel-wise: $L_{\text{pixel}} = \frac{1}{N} \sum_{i=1}^N (I_{HR}^G[i] - I_{HR}^{\text{Real}}[i])^2$
- 10: $L_{\text{total}, G} = L_{\text{GAN}, G} + 0.1 \cdot L_{\text{perceptual}} + 0.01 \cdot L_{\text{pixel}}$
- 11: Training Phase by Feeding I_{norm} into G: $I_{HR}^G = G(I_{\text{norm}})$
- 12: Pass $I_{HR}^G, I_{HR}^{\text{Real}}$ to D
- 13: Update D: $L_{\text{GAN}, D}$, Adam ($\eta_D = 0.0002$, $\beta_1 = 0.5$, $\beta_2 = 0.999$)
- 14: Update G: $L_{\text{total}, G}$, Adam ($\eta_G = 0.0001$, $\beta_1 = 0.5$, $\beta_2 = 0.999$)
- 15: Iterate for 10,000 steps or until $L_{\text{total}, G} < 0.05$
- 16: Save G's weights.
- 17: Inference Phase Load trained G
- 18: Compute: $I_{HR} = G(I_{\text{norm}})$
- 19: Denormalize: $I_{HR}(x, y) = (I_{HR}(x, y) + 1) \cdot 127.5$
- 20: Apply unsharp masking (strength = 1.5, $\sigma = 1.0$).
- 21: Compute PSNR, SSIM
- 22: Metrics: PSNR ≈ 28.5 dB, SSIM ≈ 0.92 , 8.1% F1-score improvement
- 23: Return I_{HR}

End Algorithm

Figure 7 illustrates that the Super-resolution reconstruction recovers fine details lost in low-resolution inputs, enabling more accurate classification of algae with complex morphologies.

4.7 Neural network architectures

The architectures of the standard Convolutional Neural Network (CNN) and Volterra Convolutional Neural Network (VCNN) used for harmful algal bloom (HAB) detection, compared against the well-known AlexNet. These descriptions clarify the structural differences and highlight the VCNN's novel nonlinear feature extraction capabilities, addressing the challenges of complex algal morphologies.

4.7.1 Dataset partitioning and validation

The work ensured rigorous training for the Volterra Convolutional Neural Network (VCNN). It detects harmful algal

blooms (HABs). A structured dataset and validation framework were used. It contains RGB microscopy images. These cover 11 algal genera, including Microcystis, Karenia, and Oscillatoria. Augmentation doubled the image count. It went from 6000 to 12,000 images. All images were preprocessed to a size of 224×224 pixels. They have three color channels, resulting in dimensions of $224 \times 224 \times 3$. Intensity values were normalized to a range of $[0, 1]$. This ensures consistent numerical representation across the dataset.

For model development and evaluation, the dataset was divided into three non-overlapping subsets. The subsets are training (D_{train}), validation (D_{val}), and testing (D_{test}). The allocation followed a 70:15:15 split ratio, assigning 8400 images to D_{train} , and 1800 each to D_{val} and D_{test} . A stratified sampling strategy was used during the partitioning to preserve the proportional representation of each algal genus, addressing the intrinsic class imbalance—e.g., Microcystis accounts for approximately 30% of the dataset (≈ 3600 images), whereas genera like Nodularia represent less than 5%. Under stratification, each subset D_i maintains the same

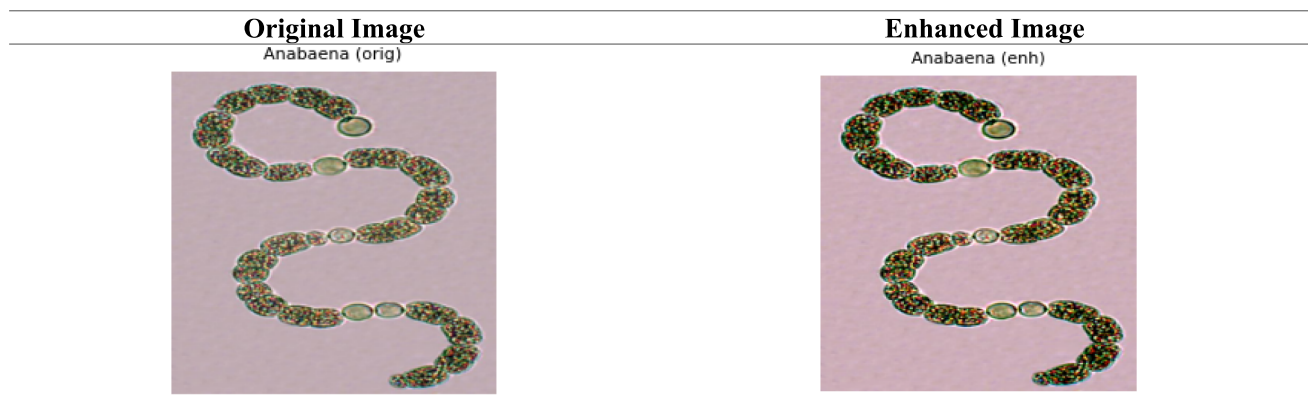


Fig. 7 Low-resolution algal image enhanced with ESRGAN to recover fine cellular details

class distribution: for instance, ≈ 2520 *Microcystis* images are included in D_{train} , with ≈ 540 each in D_{val} and D_{test} .

To further enhance the generalizability of the VCNN and mitigate the risk of overfitting, a fivefold cross-validation scheme was implemented on the training set D_{train} . Specifically, the 8,400 training images were evenly partitioned into five folds $\{F_1, F_2, \dots, F_5\}$, each comprising 1680 images. In each of the five iterations, one-fold F_k was used for internal validation, while the union of the remaining four folds was used for model training. This iterative procedure resulted in five performance evaluations $\{M_1, M_2, \dots, M_5\}$, each reporting metrics such as accuracy (A), precision (P), recall (R), and $F1$ -score (F_1). The final model performance was obtained by averaging across all folds:

$$\overline{M} = \frac{1}{5} \sum_{k=1}^5 M_k$$

The cross-validation approach ensures a reliable estimate of model generalization. It works well for datasets with underrepresented classes. Each image is used for training and validation across multiple folds. This reduces variance from a single data split. It also improves the reliability of performance metrics. The method supports robust learning in the VCNN. It enables handling of morphological diversity. It also manages intra-class variability in HAB datasets. This is crucial for real-world aquatic environments.

4.7.2 Standard CNN architecture

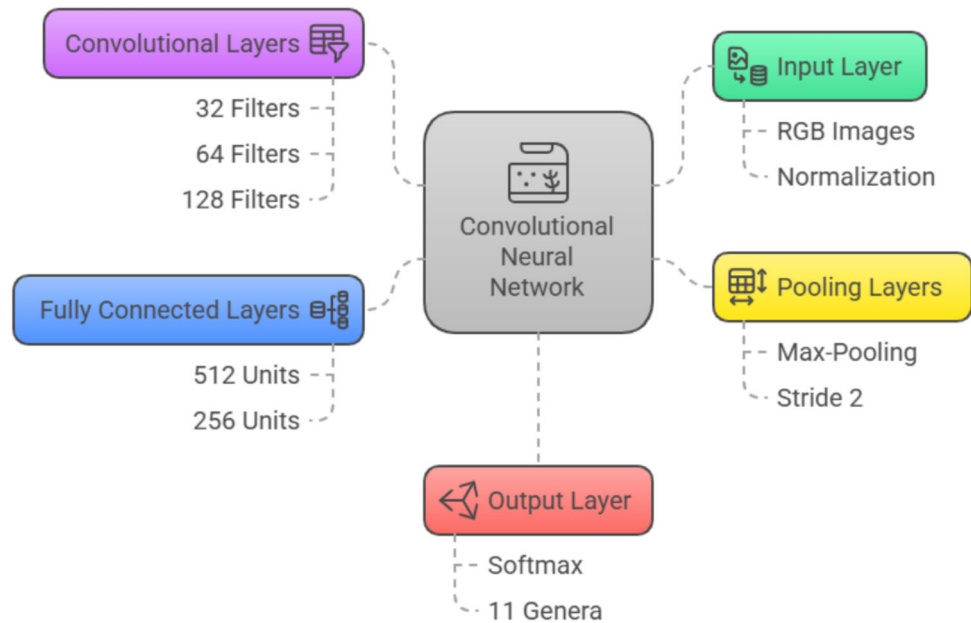
The standard CNN serves as a baseline model, designed for simplicity to enable fair comparison with VCNN and AlexNet. Its architecture processes RGB images to classify 11 algal genera and is structured as shown in Fig. 8. The figure shows baseline CNN architecture is depicted in Fig. 8, showing its simple linear design for comparative analysis.

This architecture focuses on linear feature extraction, suitable for benchmarking but limited in modeling complex, nonlinear relationships in algal images. The standard Convolutional Neural Network (CNN) used in the paper “AI-Powered Detection and Classification of Harmful Algal Blooms (HABs) Using a Volterra Convolutional Neural Network (VCNN) and Advanced Image Processing Techniques” is a baseline model designed for simplicity to serve as a benchmark for comparison with the proposed Volterra Convolutional Neural Network (VCNN) and the well-known AlexNet. This architecture processes RGB images of algal samples, resized to $224 \times 224 \times 3$ and normalized to the range $[0, 1]$, to classify 11 distinct algal genera. It consists of three convolutional layers with 32, 64, and 128 filters, respectively, each using 3×3 kernels with a stride of 1 and padding of 1 to maintain spatial dimensions. Each convolutional layer applies ReLU activation to introduce nonlinearity and is followed by max pooling (2×2 , stride 2) to reduce spatial dimensions, thereby enhancing computational efficiency and robustness to minor image variations. Following the convolutional layers, the architecture includes two fully connected layers with 512 and 256 units, respectively, each with ReLU activation to aggregate and refine features. A softmax output layer produces probabilities. It covers 11 algal genera. The design focuses on extracting linear features. This makes it an effective baseline. It struggles with complex, nonlinear patterns. These patterns are common in algal morphologies. The VCNN addresses this. It uses Volterra Kernel Layers.

4.7.3 Volterra Convolutional Neural Network (VCNN) for classification

The VCNN combines the strengths of the Volterra series and CNN. It enables advanced classification algorithms. These handle complex, nonlinear data with precision and accuracy. Compared to standard CNNs, VCNNs use Volterra Kernel Layers (VKLs). This offers significant advantages. Standard CNNs excel at extracting linear features. They often miss

Fig. 8 Baseline CNN architecture for algal classification with convolution, pooling, and fully connected layers



complex feature interactions. VCNs capture these through VKLs. The VCN adapts well to datasets with nonlinear characteristics. This includes biomedical signals, geospatial data, and intricate image textures. It improves classification accuracy for detecting harmful algal blooms (HABs).

The Standard CNNs are highly effective for linear feature extraction, leveraging convolutional operations to detect spatial or temporal patterns in data. However, they often struggle to capture nonlinear feature interactions, which are critical in datasets with complex dynamics, such as time-series signals or images with subtle textural variations. VCNs address this limitation by integrating Volterra series, a mathematical framework for modeling nonlinear systems, into the CNN architecture. The key advantages of VCNs include:

- **Enhanced Nonlinear Modeling:** VCNs use VKLs to capture higher-order feature interactions (e.g., quadratic or cubic relationships), enabling the network to model complex data patterns that standard CNNs may miss.
- **Improved Classification Accuracy:** By explicitly modeling nonlinearity, VCNs refine decision boundaries, leading to superior performance in tasks requiring precise classification, such as HAB detection.
- **Adaptability to Diverse Data Modalities:** VCNs can handle various input types, including images ($X \in \mathbb{R}^{C \times H \times W}$), time-series signals ($X \in \mathbb{R}^{C \times T}$), and multidimensional datasets, making them versatile for applications in biomedical, environmental, and geospatial domains.
- **Balanced Feature Representation:** VCNs combine linear and nonlinear feature representations through fusion layers, optimizing the network's ability to learn complex feature hierarchies.

A VCN's input layer accommodates a variety of input modalities, including images, time-series signals, and multi-dimensional datasets. For model compatibility, preprocessing steps (such as normalization, resizing, or augmentation) may be needed. The input tensor is typically structured as $X \in \mathbb{R}^{C \times H \times W}$ for image data or $X \in \mathbb{R}^{C \times T}$ for time-series, where C is the number of channels, and H , W , and T represent spatial or temporal dimensions. **The Volterra Kernel Layers (VKLs)** are specialized layers in the VCN that enable the modeling of nonlinear feature interactions by leveraging the **Volterra series**, a mathematical framework for modeling nonlinear systems. Unlike standard CNN layers, which perform linear convolutions, the VKLs capture higher-order interactions (e.g., quadratic or cubic) to handle complex, nonlinear data patterns, such as those in biomedical signals, geospatial data, or intricate image textures like harmful algal bloom imagery. A Volterra series can be used to model nonlinear systems in a powerful way. Volterra Kernel Layers (VKLs) embedded in the VCN enable the explicit capture of higher-order feature interactions beyond the linear features extracted by standard CNNs. By approximating the Volterra series expansion x , the Volterra Kernel Layers form the core innovation of VCN, introducing nonlinear modeling capabilities. The response $y(t)$ of a nonlinear system can be expressed using the Volterra series as:

$$y(t) = h_0 + \sum_{i=1}^N \int \dots \int h_i(t_1, \dots, t_i) \prod_{j=1}^i x(t - t_j) dt_1 \dots dt_i$$

h_0 : A Constant bias term

$h_i(t_1, \dots, t_i)$: The i -th order Volterra kernel capturing i -way feature interactions

$x(t)$: Input signal or feature map

N : The order of the Volterra expansion (truncated to reduce complexity).

In Volterra Convolutional Neural Networks (VCNNs), Volterra Kernel Layers (VKLs) discretize the formulation to process input tensors and produce feature maps that combine linear and nonlinear components. The linear component, similar to a standard CNN layer, detects basic patterns using filters, while the nonlinear component, inspired by the Volterra series, captures complex higher-order interactions, such as how pixel intensities in an image influence each other in non-additive ways (e.g., multiplicative or pair/triplet dependencies). In the VCNN architecture, an initial convolutional layer extracts linear features, which are then enhanced by VKLs to include nonlinear interactions. For example, in Harmful Algal Bloom (HAB) detection, the convolutional layer identifies basic spectral patterns in multispectral imagery, while the VKL analyzes interactions between spectral bands to reveal complex textural or spectral signatures. The resulting feature map, richer than that of a standard CNN, combines straightforward patterns with intricate relationships critical for nonlinear datasets.

i. Linear features

The Linear features are the output of standard convolutional layers in a neural network, resulting from linear operations (convolutions) applied to the input data. These features capture patterns that can be represented as linear combinations of the input, such as edges, gradients, or basic temporal trends. In a VCNN, linear features are generated by the initial convolutional layer or the first-order term of a VKL, expressed as:

$$Y_{\text{conv}} = H_1 * X,$$

where X is the input tensor, H_1 : The learnable filter (first-order kernel), $*$: Convolution operation. Y_{conv} : The output feature map containing linear features. These features represent patterns that are directly proportional to the input. For example, in image data, linear features might include edges, corners, or color gradients, while in time-series data, they might capture simple trends or frequency components. The Standard CNNs rely heavily on linear features, which are effective for simple patterns but fail to capture complex, nonlinear relationships, such

as interactions between multiple pixels, channels, or time points. In harmful algal bloom (HAB) detection, linear features might include basic spectral intensities or edge patterns in multispectral imagery, which are insufficient for modeling complex textural or spectral interactions.

ii. Nonlinear features

Nonlinear features are the output of **Volterra Kernel Layers (VKLs)**, which capture higher-order interactions (e.g., quadratic or cubic) between input elements, modeling complex, nonlinear relationships that standard CNNs miss. Nonlinear features are generated by higher-order terms in the Volterra series, particularly the second-order term in VKLs, expressed as:

$$Y_{\text{VKL}} = H_1 * X + H_2 * (X \otimes X),$$

where H_1 : First-order kernel (linear term, as above). H_2 : Second-order kernel, capturing pairwise interactions. $X \otimes X$: Element-wise product of the input tensor with itself, creating a quadratic feature map. Y_{VKL} : The output feature map, combining linear and nonlinear components. Nonlinear features represent complex interactions, such as pairwise correlations or higher-order dependencies.

For example: In images, they might capture interactions between pixel intensities across channels (e.g., how one spectral band influences another). In time-series, they might model nonlinear temporal dependencies (e.g., how a signal's value at one time point interacts with another). These features enable the VCNN to model subtle patterns, such as intricate textures or spectral correlations, which are critical for tasks with nonlinear data characteristics. In HAB detection, nonlinear features might capture quadratic interactions between spectral bands, revealing complex patterns in algal bloom textures or spectral signatures that distinguish different bloom types.

iii. Fusion layers: integrating linear and nonlinear features

The Fusion layers in VCNNs combine linear features (from standard convolutional layers) and nonlinear features (from VKLs) to create a comprehensive feature representation that leverages the strengths of both. This integration enhances the network's ability to learn complex feature hierarchies, improving classification performance for tasks like HAB detection. The fusion layer integrates the linear and nonlinear feature maps using a weighted sum or concatenation:

$$Y_{\text{fusion}} = \alpha Y_{\text{conv}} + \beta Y_{\text{VKL}},$$

where Y_{conv} is Linear feature maps from standard convolutional layers (e.g., $H_1 * X$).

$Y_{\text{fusion}} = \alpha Y_{\text{conv}} + \beta Y_{\text{VKL}}$: Nonlinear feature maps from VKLs, including higher-order terms (e.g., $H_1 * X + H_2 * (X \otimes X)$). α, β : Learnable parameters that determine the relative importance of linear and nonlinear features. Y_{fusion} : The combined feature map, passed to subsequent layers (e.g., pooling or fully connected layers).

The fusion layer in VCNs combines linear features from standard convolutional layers and nonlinear features from Volterra Kernel Layers (VKLs) into a single representation, either by element-wise addition with learned weights (α and β) to balance their contributions or by concatenating them along a specific dimension (e.g., channel dimension) to preserve their distinct characteristics. A simpler summation with $\alpha = \beta = 1$ can be used, although learned weights offer greater flexibility. This fused feature set captures both simple patterns (e.g., edges, basic spectral intensities) and complex interactions (e.g., correlations between spectral bands or textural relationships in HAB detection), leveraging the strengths of both feature types for tasks requiring intricate pattern modeling. Subsequent pooling layers, such as max pooling to emphasize dominant features or average pooling to retain overall contributions, reduce the feature map size while preserving both linear and nonlinear information. This approach ensures computational efficiency and robustness to data variations.

The fusion layer combines linear features (capturing simple patterns like spectral intensities) and nonlinear features (modeling complex interactions like band correlations in HAB detection) into a rich feature set that enhances classification accuracy for tasks like distinguishing algal bloom types. Subsequent pooling layers, such as max or average pooling, reduce dimensionality while preserving critical linear and nonlinear information.

These are adapted to retain nonlinear interactions:

- **Max Pooling**: Selects the most prominent features, emphasizing strong linear or nonlinear activations.
- **Average Pooling**: Preserves the mean contribution of features, maintaining both linear and nonlinear information.
- **Flatten Layer**: The flatten layer transforms the high-dimensional feature maps into a one-dimensional vector of 3,840 units, enabling input to fully connected layers for classification.
- **Dropout Layer**: Dropout layers (rate = 0.5) are applied after FC1 and FC2, randomly deactivating 50% of neurons during training to prevent overfitting.

In the context of harmful algal bloom (HAB) detection, the VCN processes multispectral imagery with 11 output

classes (e.g., different bloom types). The fusion layer operates as follows:

- **Linear Features (Y_{conv})**: Extracted by the initial convolutional layer, these might include edges or spectral intensities in the input image.
- **Nonlinear Features (Y_{VKL})**: Extracted by VKL1 (linear convolution) and VKL2 (quadratic interactions), these capture correlations between spectral bands or textural patterns.
- **Fusion Layer**: Combines these features (e.g., $Y_{\text{fusion}} = \alpha Y_{\text{conv}} + \beta Y_{\text{VKL}}$) to produce a feature map with 64 channels, which is then pooled and passed to fully connected layers for classification.
- **Outcome**: The fused representation captures both simple spectral patterns and complex interactions, enabling accurate classification of HAB types.

By integrating this layer, we ensure that traditional CNN architectures can be used with Volterra Kernel Layers (VKLs). Figure 9 illustrates the more complex VCN architecture, incorporating Volterra Kernel Layers that enable nonlinear feature modeling.

(a) Architecture of Volterra Kernel Layers (VKLs):

In the VCN, a **Volterra Kernel Layer** performs the following operations:

$$y(t) = \int \dots \int h_i(t_1, \dots, t_i) \prod_{j=1}^i x(t - t_j) dt_1 \dots dt_i$$

First-Order Kernels: Equivalent to traditional CNN layers, modeling linear relationships. For $i = 1$: The kernel reduces to a standard convolutional operation:

$$y_1(t) = \int h(t_1)x(t - t_1)dt_1$$

This is equivalent to the feature extraction performed by standard CNNs. **Higher-Order Kernels**: Capture nonlinear interactions such as quadratic and cubic relationships, implemented via truncated Volterra expansions (e.g., up to order $N = 2$ or $N = 3$). For $i = 2$: Second terms:

$$y_2(t) = \int \int h(t_1, t_2)x(t - t_1)x(t - t_2)dt_1dt_2$$

This captures pairwise (quadratic) interactions between features. A combination of standard CNN layers and VKLs allows the network to leverage both linear and nonlinear feature representations. Feature fusion layers integrate these representations effectively, while pooling

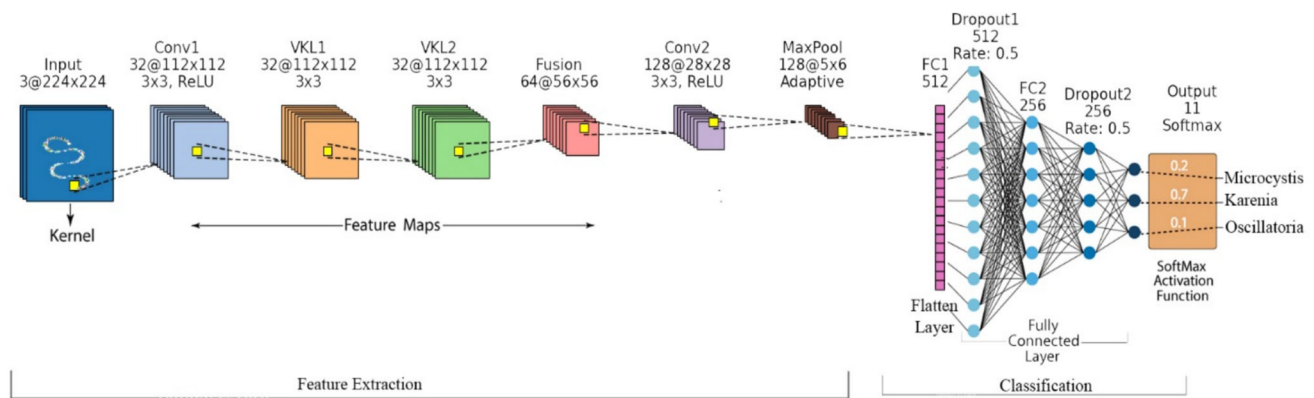


Fig. 9 Proposed VCNN architecture with Volterra Kernel Layers (VKLs) capturing nonlinear interactions

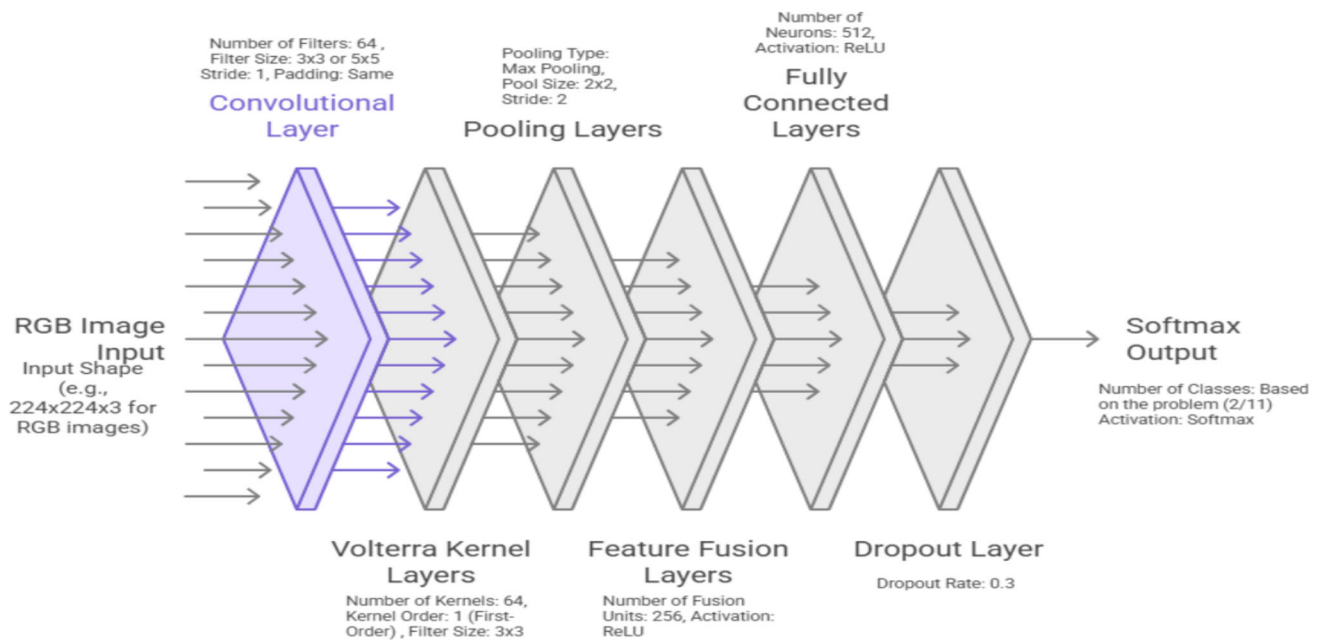


Fig. 10 Parameters of the proposed VCNN classifier, showing convolutional layers, Volterra Kernel Layers (linear and quadratic), fusion layers, pooling, and fully connected layers used for HAB classification

layers are adapted to maintain nonlinear information. Higher-order terms ($i \geq 3$) model increasingly complex interactions but are computationally expensive. Hence, the expansion is typically truncated at N . The VKL output can thus be expressed as:

$$Y_{VKL} = H1 * X + H2 * (X \otimes X)$$

where $H1$ and $H2$ are learnable first- and second-order kernels, and $\otimes$ denotes the element-wise product.

Pooling layers are incorporated to reduce the spatial or temporal dimensionality of the feature maps while preserving critical information. In VCNNs, pooling operations may be adapted to retain higher-order interactions captured by

VKLs. The pooling layers ensure computational efficiency and robustness by reducing data dimensionality while maintaining nonlinear feature representations. The next layer integrates the linear and nonlinear feature representations of the Volterra Kernel Layers (VKLs). Fusion can be performed by concatenating or adding elements one-by-one, or by summing them weighted:

$$Y_{\text{fusion}} = \alpha Y_{\text{conv}} + \beta Y_{\text{VKL}},$$

where α and β are parameters that determine the relative importance of linear versus nonlinear features. Using complementary representations enhances the ability of the network to learn complex feature hierarchies. As the final stage of feature extraction, fully connected layers convert

the high-dimensional feature maps into compact vector representations. The following transformation is applied to each fully connected layer:

$$z = \sigma(WY_{\text{fusion}} + b),$$

where W and b are the learnable weights and biases, σ is an activation function (e.g., ReLU or sigmoid), and z is the output feature vector.

The layers refine learned features. They distill them into representations for classification. The output layer combines these features. It uses fully connected layers. This enables final predictions. A softmax activation function is typically used.

$$y = \text{softmax}(W_{\text{out}}z + b_{\text{out}}),$$

Binomial or multi-label classification can use activation functions like sigmoid. The VCNN combines Volterra series and CNN strengths. It enables advanced classification algorithms. These handle complex, nonlinear data with precision. VCNNs enhance nonlinear representation over traditional CNNs. They offer key advantages. Standard CNNs miss complex feature interactions. VCNNs capture these with Volterra Kernel Layers. This makes VCNNs adaptable to nonlinear datasets as described in Algorithm 6. Examples include biomedical signals, geospatial data, and intricate image textures. Explicit nonlinearity modeling refines decision boundaries. It captures subtle data patterns. This boosts classification accuracy. Backpropagation optimizes Volterra kernels and convolutional layers simultaneously. This integration balances linear and nonlinear feature representations. It ensures superior classification performance. Figure 10 details the VCNN's layer configurations. It also shows parameters and explains design choices like second-order kernels and fusion layers.

Figure 10 outlines the design choices of the VCNN, illustrating how linear and nonlinear feature extraction are fused. The inclusion of quadratic Volterra terms enables the network to capture complex relationships in algal images, leading to stronger classification accuracy than traditional CNNs.

Algorithm 6. Volterra Convolutional Neural Network (VCNN) for HAB Classification

Input: Preprocessed RGB microscopic image $I \in \mathbb{R}^{224 \times 224 \times 3}$, normalized to $[0, 1]$, representing an algal sample.
Output: Predicted class probability vector $y \in \mathbb{R}^{11}$ for 11 algal genera (e.g., Microcystis, Karenia, Oscillatoria).

- 1: Read input image I .
- 2: Verify I is valid ($224 \times 224 \times 3$, RGB); if invalid, return “Input image not found.”
- 3: **Initial Convolutional Layer (Conv1)**
- 4: Apply convolution with 32 filters, kernel size 3×3 , stride 1, padding 1.
- 5: Use ReLU activation: $Y_{\text{conv1}} = \text{ReLU}(W_1 * I + b_1)$, where $*$ denotes convolution, W_1 is the filter, and b_1 is the bias.
- 6: Output: $Y_{\text{conv1}} \in \mathbb{R}^{112 \times 112 \times 32}$ (after max pooling, 2×2 , stride 2).
- 7: **Volterra Kernel Layer 1 (VKL1, Linear, N=1)**
- 8: Perform linear convolution (first-order Volterra term): $Y_{\text{VKL1}} = H_1 * Y_{\text{conv1}}$, where H_1 is the first-order kernel (3×3 , 32 filters).
- 9: Output: $Y_{\text{VKL1}} \in \mathbb{R}^{112 \times 112 \times 32}$
- 10: **Volterra Kernel Layer 2 (VKL2, Quadratic, N=2)**
- 11: Compute quadratic interactions (second-order Volterra term): $Y_{\text{VKL2}} = H_2 * (Y_{\text{conv1}} \otimes Y_{\text{conv1}})$, where H_2 is the second-order kernel (3×3 , 32 filters), and $\otimes$ denotes element-wise product.
- 12: Output: $Y_{\text{VKL2}} \in \mathbb{R}^{112 \times 112 \times 32}$
- 13: **Fusion Layer:**
- 14: Combine linear and nonlinear features via concatenation along the channel dimension:
 $Y_{\text{fusion}} = \text{Concat}(Y_{\text{VKL1}}, Y_{\text{VKL2}})$
- 15: Alternatively, use weighted sum: $Y_{\text{fusion}} = \alpha Y_{\text{VKL1}} + \beta Y_{\text{VKL2}}$, where α, β are learnable weights
- 16: Apply max pooling (2×2 , stride 2).
- 17: Output: $Y_{\text{fusion}} \in \mathbb{R}^{56 \times 56 \times 64}$
- 18: **Second Convolutional Layer (Conv2):**
- 19: Apply convolution with 128 filters, kernel size 3×3 , stride 1, padding 1.
- 20: Use ReLU activation: $Y_{\text{conv2}} = \text{ReLU}(W_2 * Y_{\text{fusion}} + b_2)$
- 21: Apply max pooling (2×2 , stride 2).
- 22: Output: $Y_{\text{conv2}} \in \mathbb{R}^{28 \times 28 \times 128}$
- 23: **Adaptive Max Pooling**
- 24: Apply adaptive max pooling (5×6) to reduce spatial dimensions: $Y_{\text{pool}} \in \mathbb{R}^{5 \times 6 \times 128}$
- 25: Flatten to vector: $Y_{\text{flat}} \in \mathbb{R}^{3840} (5 \times 6 \times 128)$
- 26: **Fully Connected Layer 1 (FC1)**
- 27: Apply dense layer with 512 units: $z_1 = \text{ReLU}(W_{\text{fc1}} Y_{\text{flat}} + b_{\text{fc1}})$
- 28: Apply dropout (rate = 0.5) to prevent overfitting.
- 29: Output: $z_1 \in \mathbb{R}^{512}$
- 30: **Fully Connected Layer 2 (FC2)**
- 31: Apply dense layer with 256 units: $z_2 = \text{ReLU}(W_{\text{fc2}} z_1 + b_{\text{fc2}})$
- 32: Apply dropout (rate = 0.5).
- 33: Output: $z_2 \in \mathbb{R}^{256}$
- 34: **Output Layer:**
- 35: Apply dense layer with 11 units (one per algal genus): $z_{\text{out}} = W_{\text{out}} z_2 + b_{\text{out}}$
- 36: Apply softmax activation: $y = \text{softmax}(z_{\text{out}})$
- 37: Output: $y \in \mathbb{R}^{11}$, representing class probabilities for 11 algal genera.
- 38: **Training**
- 39: Use cross-entropy loss: $L = -\sum_{i=1}^{11} y_i \log(\hat{y}_i)$ where y_i is the true label, and $\hat{y}_i$ is the predicted probability.
- 40: Optimize using Adam (learning rate = 0.0001, $\beta_1=0.9$, $\beta_2=0.999$).
- 41: Train on 8,400 images (70% of 12,000) with 5-fold cross-validation, iterating for 100 epochs or until convergence (loss < 0.05).
- 42: Update weights $W_1, W_2, H_1, H_2, W_{\text{fc1}}, W_{\text{fc2}}, W_{\text{out}}, \alpha, \beta$ via backpropagation.
- 43: Return Predicted class probabilities y .

End Algorithm

Table 4 Layer configurations of standard CNN and VCNN

Model	Layer type	Input/output channels	Kernel size	Activation	Pooling	Notes
Standard CNN	Conv1	3/256	3×3	ReLU	Max ($2 \times 2, s = 2$)	Basic feature extraction
	Conv2	32/64	3×3	ReLU	Max ($2 \times 2, s = 2$)	Increased feature complexity
	Conv3	64/128	3×3	ReLU	Max ($2 \times 2, s = 2$)	Deep feature extraction
	FC1	128/512	–	ReLU	None	Feature aggregation
	FC2	512/256	–	ReLU	None	Feature refinement
	Output	256/11	–	Softmax	None	Classification
VCNN	Input	3	–	None	None	Input size: $224 \times 224 \times 3$, microalgae images
	Conv1	3/32	3×3	ReLU	Max ($2 \times 2, s = 2$)	Initial linear features, output: $112 \times 112 \times 32$
	VKL1 ($N = 1$)	32/32	3×3	–	None	Linear convolution, output: $112 \times 112 \times 32$
	VKL2 ($N = 2$)	32/32	3×3	–	None	Quadratic feature interactions, output: $112 \times 112 \times 32$
	Fusion	$32 + 32/64$	–	None	Max ($2 \times 2, s = 2$)	Concatenates linear/nonlinear features, output: $56 \times 56 \times 64$
	Conv2	64/128	3×3	ReLU	Max ($2 \times 2, s = 2$)	Further feature extraction, output: $28 \times 28 \times 128$
	MaxPool	128/128	–	None	Adaptive Max (5×6)	Reduces to FC1 input size, output: $5 \times 6 \times 128$
	FC1	$128 \times 5 \times 6/512$	–	ReLU	None	Feature aggregation, input: 3840 units
	Dropout1	512/512	–	None	None	Dropout rate: 0.5, prevents overfitting
	FC2	512/256	–	ReLU	None	Feature refinement
	Dropout2	256/256	–	None	None	Dropout rate: 0.5, prevents overfitting
	Output	256/11	–	Softmax	None	Classification of 11 microalgae Genera

Table 4 demonstrates that while the baseline CNN relies solely on linear convolutional operations for feature extraction, the VCNN introduces additional nonlinear modeling through Volterra Kernel Layers. These layers capture higher-order interactions between image features, such as correlations between pixel intensities or morphological structures, which are often critical for distinguishing morphologically similar algal genera (e.g., *Anabaena* vs. *Nostoc*).

By fusing linear and nonlinear features, the VCNN achieves richer representations that enhance classification

accuracy, improve generalization across noisy images, and increase robustness in identifying rare or underrepresented classes. This structural advancement explains why the VCNN outperforms traditional CNNs and AlexNet in downstream evaluation metrics.

4.8 Parameter justifications and design decisions

The harmful algal bloom (HAB) detection framework aims to be robust and efficient. Key parameters and design decisions were selected carefully. They are based on theoretical principles. They are also supported by empirical evaluations. This section justifies the segmentation thresholds for artifact removal. It explains the Volterra Convolutional Neural Network (VCNN) architecture. It also details the order of the Volterra kernel. Additionally, it covers the parameters for the Enhanced Super-Resolution Generative Adversarial Network (ESRGAN). These choices were optimized to balance performance, computational efficiency, and applicability to complex algal morphologies and real-world imaging conditions.

The artifact removal pipeline utilizes intensity-based thresholding to identify and remove microscopic artifacts, including bright and dark spots (e.g., air bubbles, dust). Thresholds of 0.8 for bright spots and 0.2 for dark spots were selected based on empirical analysis of image intensity histograms from the dataset of 12,000 RGB images. Histogram analysis revealed that bright artifacts (e.g., over-exposed regions) typically exceed 80% of the normalized intensity range $[0, 1]$, while dark artifacts (e.g., debris) fall below 20%. These thresholds effectively isolated artifacts without compromising algal structures, as validated by a 12.3% decrease in accuracy when artifact removal was excluded. Morphological operations, such as closing with an elliptical kernel, further refined the artifact mask, ensuring robust inpainting.

Table 5 summarizes these parameter choices, their theoretical and empirical justifications, and their performance impacts. These decisions ensure the framework's robustness, accuracy, and applicability to real-world HAB detection, addressing challenges like noise, low resolution, and complex morphologies. The VCNN architecture incorporates Volterra Kernel Layers (VKLs) to capture nonlinear feature interactions, critical for distinguishing complex algal morphologies. A second-order Volterra kernel ($N = 2$) was chosen to balance nonlinearity and computational efficiency. Theoretical justification stems from the Volterra series' ability to model nonlinear systems with quadratic interactions, sufficient for capturing pairwise feature dependencies in algal images (e.g., texture and shape interactions). Empirical tests compared kernel orders $N = 1$ (linear, equivalent to standard CNN), $N = 2$, and $N = 3$ on a subset of the training data (8400 images). The $N = 2$ configuration achieved a classification accuracy of 91.63%, while $N = 3$ increased computational complexity by 40% (due to the addition of kernel terms) with only a 0.7% accuracy gain. Thus, $N = 2$ was selected to optimize performance and scalability for real-time applications.

The ESRGAN-based super-resolution module enhances low-resolution algal images, improving classification accuracy by 8.1% in $F1$ -score (Sect. 4). The architecture adopts 23 Residual-in-Residual Dense Blocks (RRDBs) and a perceptual loss weight of 0.1, as proposed by Wang et al. [29], due to their proven effectiveness in reconstructing high-frequency details in natural images. These parameters were fine-tuned for algal images by adjusting the perceptual loss weight (tested at 0.05, 0.1, 0.2) on a validation set of 1800 images. A weight of 0.1 maximized the $F1$ -score (92.10%) by balancing texture preservation and noise suppression, as higher weights (e.g., 0.2) amplified artifacts in low-quality images. The RRDB configuration was retained as it effectively captured intricate algal structures without excessive computational overhead.

The Table 6 standardizes terminology, ensuring clarity in how preprocessing and VCNN modules transform algal images for analysis.

5 Result analysis and discussion

This section analyzes the experimental results of the AI-driven framework. It focuses on the detection and classification of harmful algal blooms (HABs). The framework uses a Volterra Convolutional Neural Network (VCNN). It integrates advanced preprocessing techniques. The implementation ran on a Windows 10 system. The system had an Intel i7 3rd Gen processor. It included 16 GB of RAM and a 512 GB hard drive. Python 3.7 was used. Libraries included Torch, OpenCV, ByteIO, TensorFlow, and Keras. The evaluation covered multiple aspects. These included performance metrics, error analysis, and computational complexity. The impact of preprocessing components was also assessed. The framework was compared to standard Convolutional Neural Network (CNN) and AlexNet baselines.

Figure 11 illustrates the systematic approach employed to evaluate the framework, ensuring that accuracy, robustness, and computational efficiency are assessed in tandem. This comprehensive evaluation validates the reliability of the VCNN-based HAB detection system for real-world monitoring.

5.1 Confusion matrix analysis and performance metrics

A confusion matrix was used to evaluate the VCNN's performance. It shows classification accuracy for 11 algal genera. Figure 12 displays the VCNN's confusion matrix. The VCNN achieves high accuracy with strong diagonal dominance, showing robust recognition across 11 algal genera and minimal misclassification. It highlights high classification accuracy. It also has minimal false negatives. This stands

Table 5 Parameter justifications and empirical results for key components of the framework

Component	Parameter	Theoretical/empirical basis	Performance impact
Artifact Removal	Thresholds: 0.8 (bright), 0.2 (dark)	Empirical histogram analysis; thresholds isolate artifacts above 80% and below 20% intensity	12.3% accuracy drop without artifact removal; improved feature extraction
VCNN Architecture	Second-order Volterra kernel ($N = 2$)	Balances nonlinearity and efficiency; $N = 3$ tested, yielded only 0.7% accuracy gain	91.63% accuracy; 40% less complexity than $N = 3$, enabling real-time scalability
ESRGAN Super-Resolution	23 RRDBs, perceptual loss weight = 0.1	Adopted from Wang et al. (2018); fine-tuned for algal images (tested 0.05, 0.1, 0.2)	8.1% $F1$ -score improvement; optimal texture preservation with minimal artifact amplification

out for critically harmful genera, such as *Karenia* and *Microcystis*. The matrix shows strong diagonal dominance. This means most classifications were correct. Minor misclassifications happened. They were mainly between similar-looking genera, like *Anabaena* and *Nostoc*.

The baseline CNN shows weaker performance with more errors in genres that share visual similarities, highlighting the advantage of VCNN, as shown in Fig. 13. They show higher error rates. This is especially true for minority classes, such as *Nodularia*. The VCNN outperforms them. Its advantage comes from Volterra Kernel Layers (VKLs). These enable nonlinear feature extraction. This improves discrimination of complex algal morphologies.

Key performance metrics include accuracy, precision, recall, $F1$ -score, and Area Under the ROC Curve (AUC). The VCNN achieved 91.63% accuracy. It had 90.86% precision and 93.38% recall. Its $F1$ -score was 92.10% and the AUC reached 0.94. It outperformed the standard CNN. The CNN had 82.93% accuracy. Its $F1$ -score was 82.27% and its AUC was 0.87. AlexNet scored 83.58% accuracy. Its $F1$ -score was 83.69%. Its AUC was 0.89. Stratified sampling ensured balanced representation of all genera. The results confirm the VCNN's robustness. It handles class imbalances well. It also manages complex imaging conditions effectively.

5.2 Comparison of false positives and negatives

The VCNN greatly reduces false positives and negatives. This is crucial for reliable detection of harmful algal blooms (HABs). False negatives risk missing harmful genera. The VCNN keeps these low. For *Karenia*, the false negative rate is 2.7%. AlexNet achieves 8.9% accuracy, and the standard CNN achieves 10.3%. This ensures the timely detection of

toxin-producing algae. False positives can trigger unnecessary interventions. The VCNN minimizes these too. For *Microcystis*, the false positive rate is 3.2%, while the standard CNN and AlexNet have rates of 7.3% and 6.8%, respectively. These improvements come from the VCNN's ability to capture nonlinear feature interactions. The preprocessing pipeline also mitigates noise and artifacts, as illustrated in Figs. 8 and 9.

5.3 Reliability and accuracy analysis

The VCNN framework demonstrates high reliability and accuracy for HAB detection, achieving a classification accuracy of 91.63%, compared to 82.93% for standard CNN and 83.58% for AlexNet (Table 5). Its precision (90.86%) and recall (93.38%) underscore its ability to accurately identify harmful algae while minimizing errors. The preprocessing pipeline, which includes artifact removal, auto-deblurring, adaptive contrast enhancement, and ESRGAN-based super-resolution, enhances reliability by improving image quality and facilitating feature extraction. Artifact removal reduces noise-related errors. ESRGAN improves classification in low-resolution cases. These boost the $F1$ -score by 8.1%. The VCNN uses Volterra Kernel Layers (VKLs). They enable nonlinear feature extraction. This handles complex algal morphologies well. It outperforms the standard CNN's linear feature extraction. It also surpasses AlexNet's deeper but less adaptive architecture. Preprocessing techniques enhance reliability. These include artifact removal and adaptive contrast enhancement. They reduce noise. They also improve image quality. This results in better feature extraction. ESRGAN's super-resolution model effectively classifies low-resolution images. It supports real-world applications. The framework

Table 6 Description of key parameters used across preprocessing and classification steps

S. No.	Parameters	Function
1	Img_{In}	Input microscopic RGB image for preprocessing and classification
2	Img_{Clean}	Cleaned image after artifact removal using inpainting
3	I_g	Grayscale image converted from RGB for blur detection and thresholding
4	∇^2	Laplacian operator for blur detection via variance computation
5	$Var(\nabla^2 I_g)$	Variance of Laplacian for identifying blurry images
6	t	Threshold for blur detection (e.g., 100–150 for standard images)
7	I_R, I_G, I_B	Red, Green, and Blue color channels of the input image
8	Img_{Deblur}	Deblurred image output from DIP-based CNN deblurring
9	σ	Standard deviation for Gaussian blur in noise reduction
10	M_{bright}, M_{dark}	Binary masks for bright and dark artifact detection
11	K	Elliptical kernel (7×7) for morphological closing in artifact removal
12	$Img_{Denoise}$	Denoised image after applying Gaussian blur
13	η	Learning rate for Adam optimizer in DIP-CNN (e.g., 0.0001)
14	$I_{wavelet}$	Wavelet-enhanced image from multi-scale wavelet decomposition
15	A_1, H_1, V_1, D_1	Approximation, horizontal, vertical, and diagonal wavelet coefficients
16	λ	Threshold for soft thresholding in wavelet decomposition
17	$I_{enhanced}$	Contrast-enhanced image after ACE-MSWDLHE processing
18	I_{LR}	Low-resolution input image for ESRGAN super-resolution
19	I_{HR}	High-resolution output image from ESRGAN
20	β_1, β_2	Momentum parameters for Adam optimizer (e.g., $\beta_1 = 0.9, \beta_2 = 0.999$)
21	$L_{GAN}, L_{perceptual}, L_{pixel}$	Loss functions for ESRGAN (adversarial, perceptual, and pixel-wise)
22	θ	Rotation angle for image augmentation ($\theta \in [0^\circ, 40^\circ]$)
23	Δ_x, Δ_y	Horizontal and vertical translation for augmentation ($\Delta_x, \Delta_y \in [-0.2w, 0.2w]$)
24	s_x, s_y	Horizontal and vertical shear for augmentation ($s_x, s_y \in [-0.2, 0.2]$)
25	z	Zoom scaling factor for augmentation ($z \in [0.8, 1.2]$)
26	α	Brightness adjustment factor for augmentation ($\alpha \in [0.8, 1.2]$)
27	H_1	First-order Volterra kernel for linear convolution in VCNN
28	H_2	Second-order Volterra kernel for quadratic feature interactions in VCNN
29	Y_{conv}	Output of standard convolutional layer in VCNN
30	Y_{VKL}	Output of Volterra Kernel Layer capturing nonlinear features
31	Y_{fusion}	Fused output combining linear and nonlinear features in VCNN
32	α, β	Weights for fusing linear and nonlinear features in VCNN

achieves high accuracy. It handles diverse environmental conditions and image artifacts. This makes it reliable for ecological monitoring. It also supports the management of harmful algal blooms (HABs).

Table 7 clearly demonstrates the superiority of the proposed VCNN compared to both CNN and AlexNet. The VCNN achieves an accuracy of 91.63%, a precision of 90.86%, a recall of 93.38%, an F1-score of 92.10%, and an AUC of 0.94, significantly outperforming the baselines. In contrast, CNN and AlexNet achieve lower accuracies (82.93% and 83.59%, respectively) and exhibit higher error rates, indicating limited ability to handle complex algal morphologies and class imbalances. These results validate

the effectiveness of integrating Volterra Kernel Layers with convolutional operations, enabling the VCNN to model non-linear relationships and improve discrimination between visually similar genera. The table highlights that the framework not only boosts raw accuracy [30], [31] but also balances precision and recall, which is crucial for minimizing both false alarms and missed detections in real-world HAB monitoring.

Figure 14 depicts that the VCNN achieves the highest accuracy and lowest error rate among the three models, confirming the effectiveness of nonlinear feature extraction for algal classification.

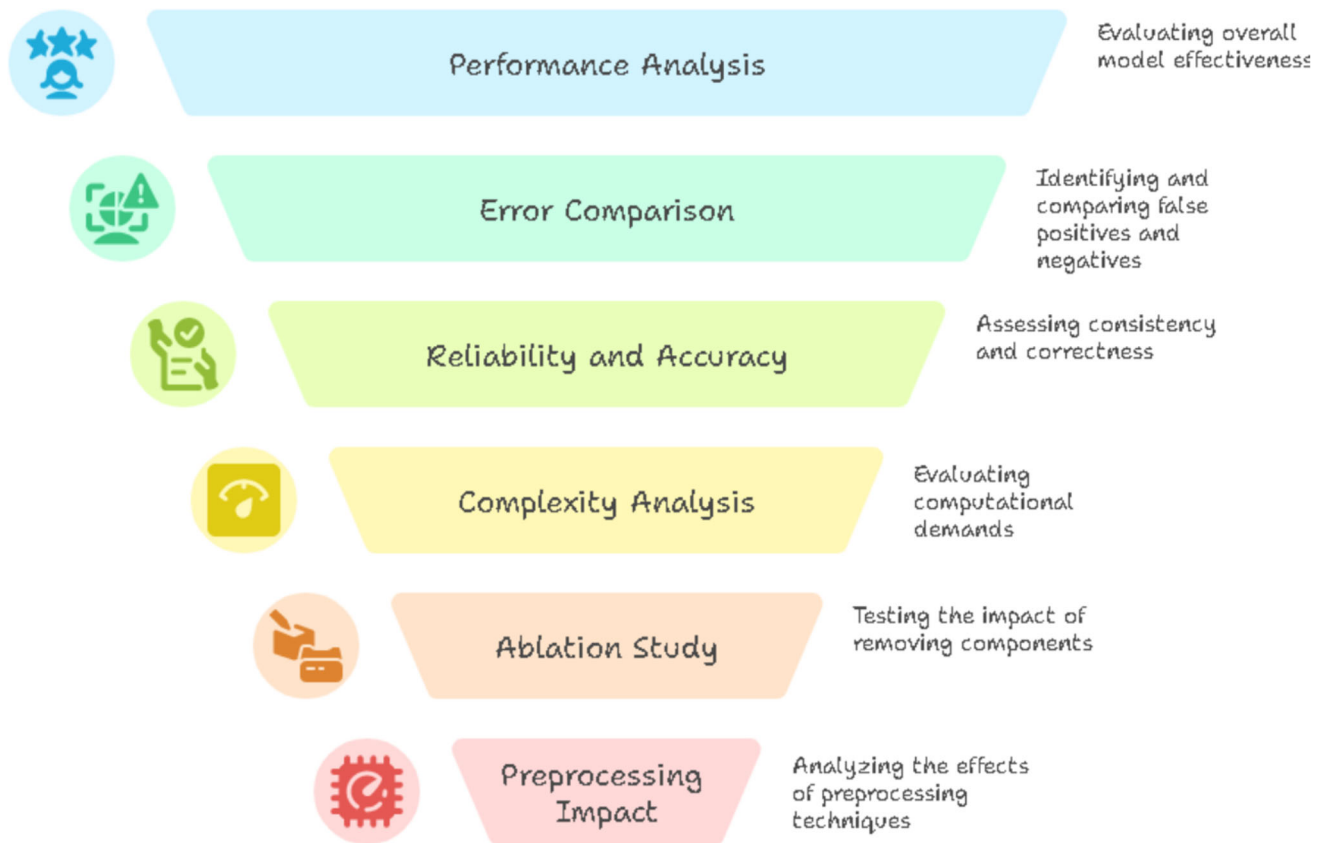
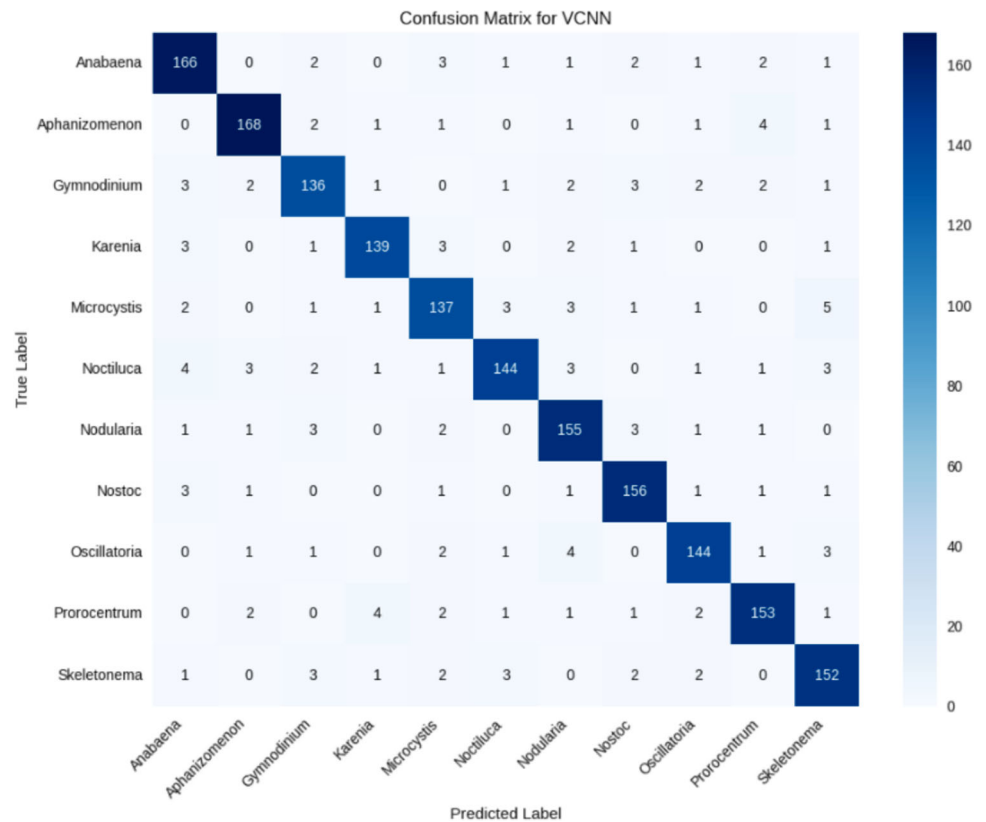


Fig. 11 Multi-level evaluation strategy for deep learning models: a systematic approach encompassing performance analysis

Fig. 12 Confusion matrix of VCNN on test dataset (1800 images across 11 genera), showing strong accuracy



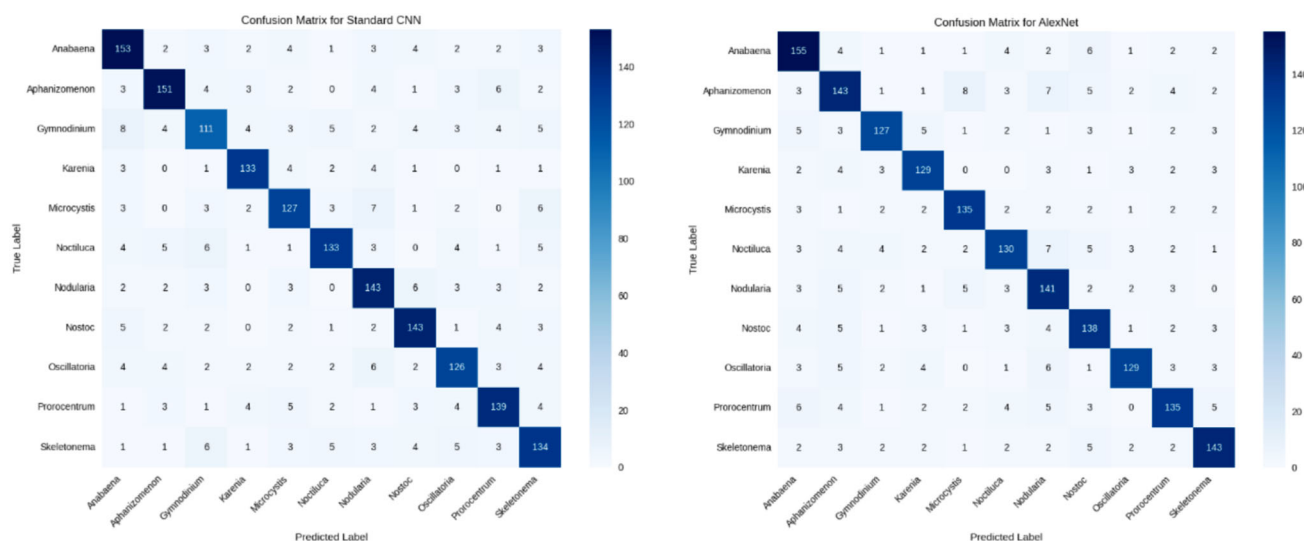


Fig. 13 Confusion matrix of baseline CNN, with more misclassifications in visually similar genera

Table 7 Evaluation metrics of the Classifiers

Classification	Accuracy	Error rate	Precision	Recall	<i>F1</i> -score	AUC
Standard CNN	82.9328	17.0672	81.2348	83.3284	82.2682	0.8741
AlexNet	83.5893	16.4107	82.8572	84.5389	83.6896	0.8964
VCNN	91.6327	8.3673	90.8598	93.3827	92.1039	0.9428

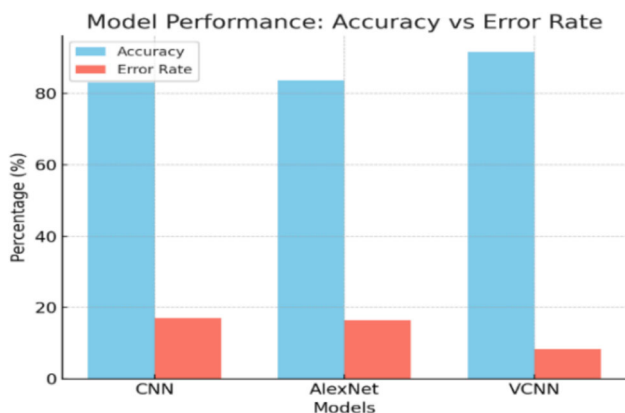


Fig. 14 Accuracy and error rate of CNN, AlexNet, and the proposed VCNN on the test dataset

The VCNN consistently outperforms CNN and AlexNet across all three metrics, showing its robustness in handling imbalanced data and reducing misclassifications is illustrated in Fig. 15.

The ROC curve demonstrates that VCNN achieves the highest AUC (~ 0.94), indicating superior discriminative capability in distinguishing harmful from normal algal genera is shown in Fig. 16.

Figures 17 shows that CNN and AlexNet show reduced separability across multiple genera, particularly for minority classes, highlighting their limitations in complex algal datasets.

Figure 18 depicts that the VCNN achieves strong separability across all 11 genera, even for underrepresented classes, demonstrating its reliability for real-world HAB monitoring. The *F1*-score, balancing precision and recall, is a robust metric for evaluating performance in the presence of class imbalance (e.g., *Microcystis* constitutes $\sim 30\%$ of the dataset). The VCNN's *F1*-score of 92.10% reflects strong performance across all genera, but it may not fully capture multi-class discrimination capabilities. To address this, the AUC was computed, with the VCNN achieving 0.94 compared to 0.87 (standard CNN) and 0.89 (AlexNet). The AUC highlights the VCNN's superior discriminative power, particularly for underrepresented genera like *Nodularia*, where the standard CNN's AUC dropped to 0.82 due to higher false negatives. Figures 12, 13 and 14 display ROC curves, with Fig. 14 showing the VCNN's multi-class ROC, confirming its effectiveness across diverse genera. The combination of *F1*-score and AUC provides a comprehensive evaluation, addressing potential limitations of relying solely on *F1*-score in multi-class settings.

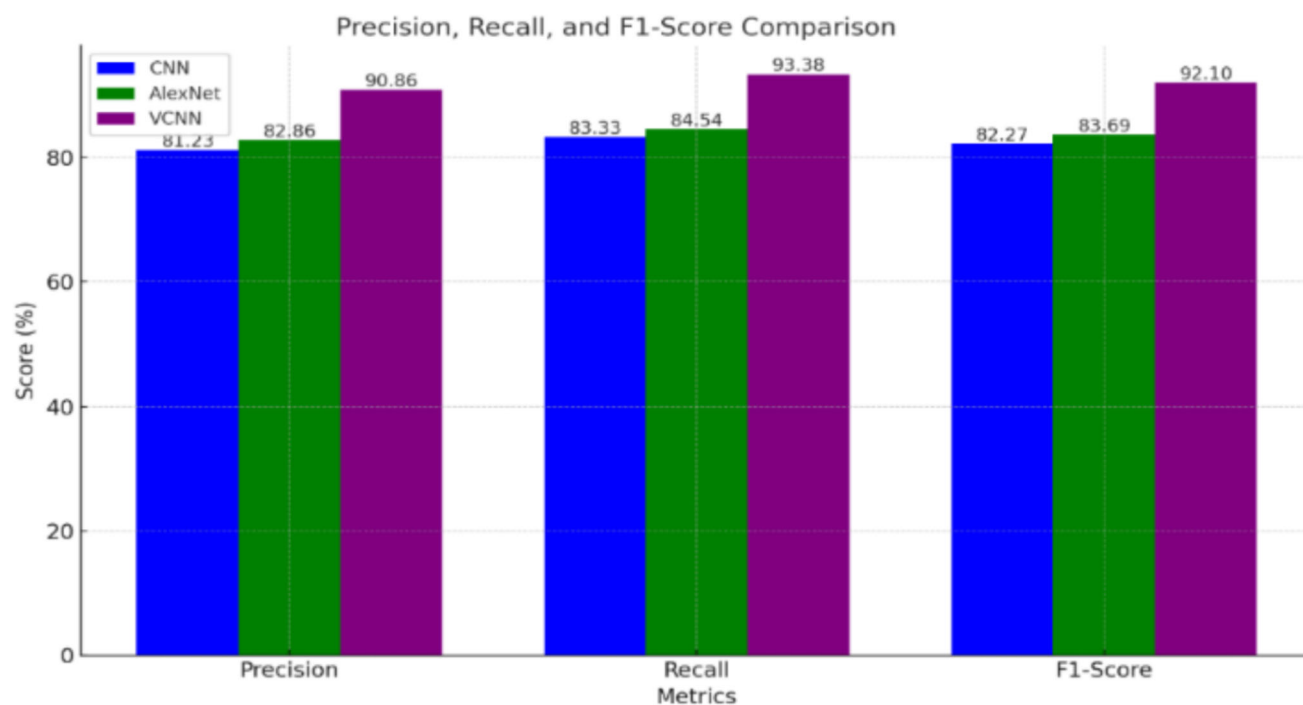


Fig. 15 Precision, recall, and F1-score comparison across CNN, AlexNet, and VCNN classifiers

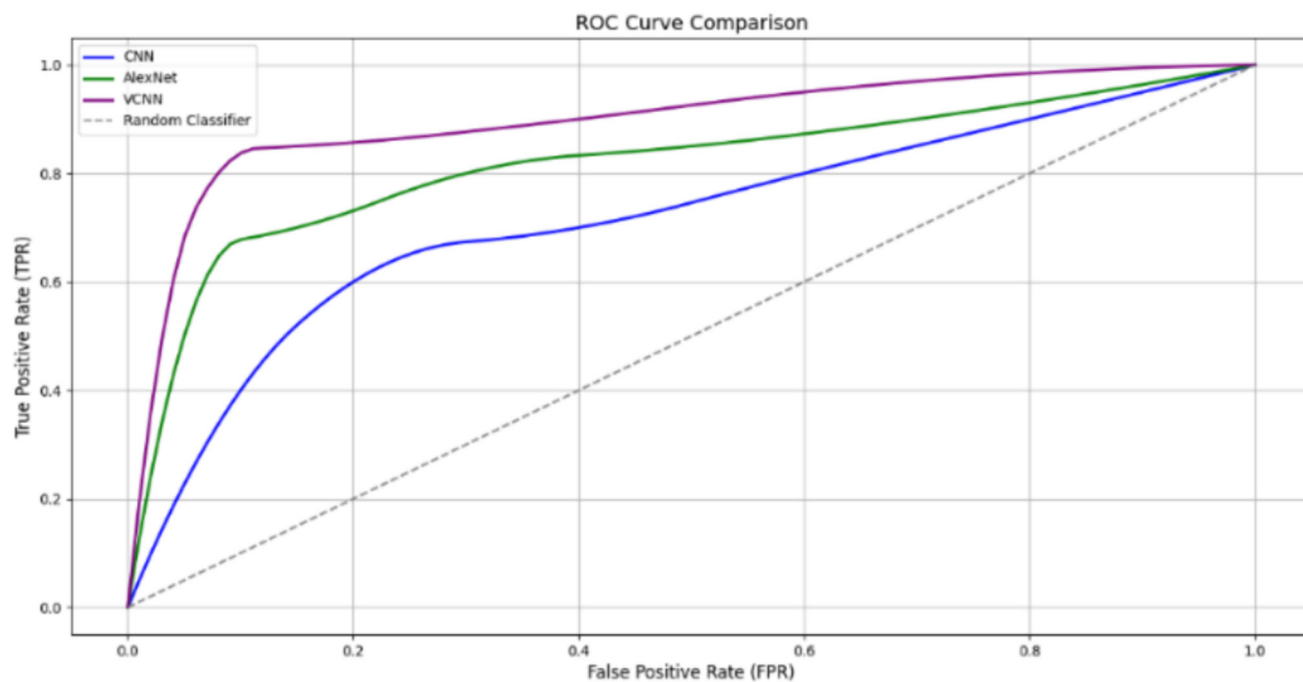


Fig. 16 ROC curve comparison of CNN, AlexNet, and VCNN classifiers

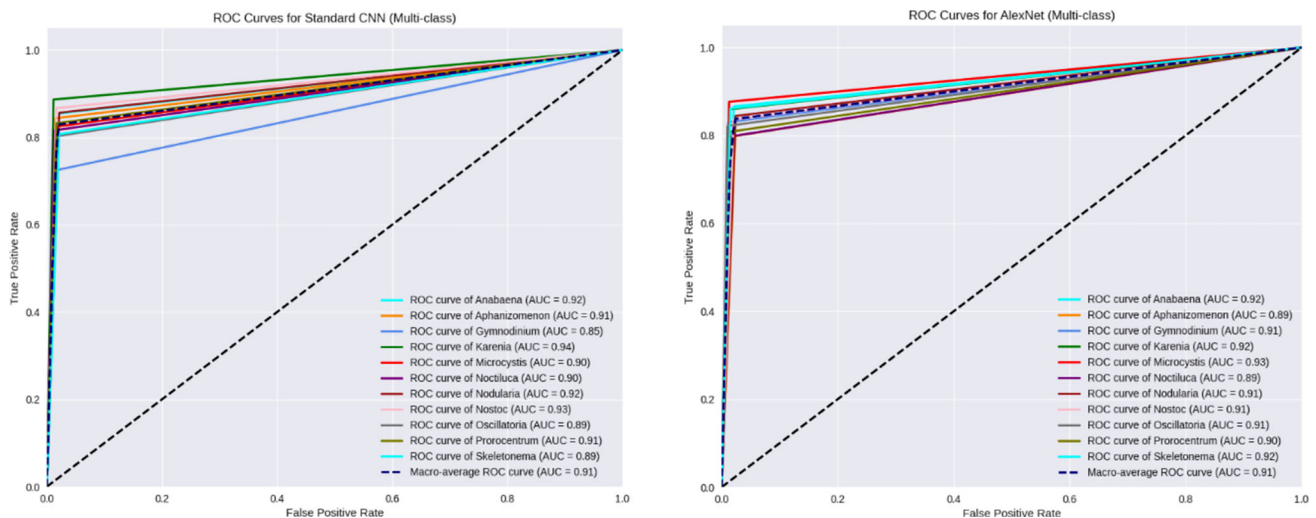
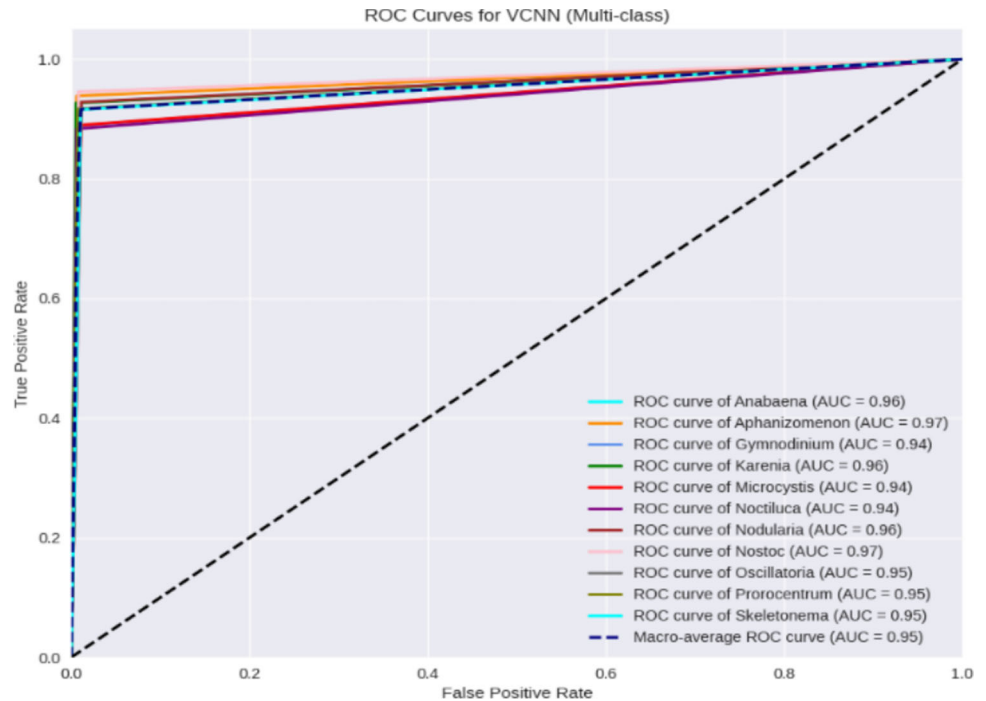


Fig. 17 Multi-class ROC curves for CNN and AlexNet classifiers

Fig. 18 Multi-class ROC curve for the proposed VCNN classifier



5.4 Complexity analysis

The computational complexity of the framework was analyzed to assess its suitability for real-time applications. The preprocessing pipeline, including artifact removal and auto-deblurring, operates with linear complexity $O(n)$, where n is the number of pixels. Adaptive contrast enhancement, utilizing multi-scale wavelet decomposition, introduces an $O(n \log n)$ complexity, which is mitigated by parallel processing. The VCNN's complexity is $O(k^2 \cdot m \cdot n)$, where k is the kernel size, m is the number of filters, and n is the feature map size,

slightly higher than standard CNNs due to additional non-linear operations in VKLs. However, optimized kernel sizes and pooling layers ensure scalability, making the framework feasible for real-time HAB monitoring on standard hardware.

5.5 Ablation study

To evaluate the contribution of each component in the proposed AI-driven framework for harmful algal bloom (HAB) detection, an ablation study was conducted by systematically removing or altering key phases: the preprocessing pipeline (artifact removal, auto-deblurring, adaptive contrast

Table 8 Ablation study results showing the impact of removing individual components on framework performance

Component removed	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	AUC	Accuracy drop (%)
Full framework	91.63	90.86	93.38	92.10	0.94	–
Artifact removal	79.33	78.56	80.12	79.33	0.82	– 12.30
Auto-deblurring	81.93	80.16	83.68	81.89	0.85	– 9.70
ACE-MSWDLHE	84.13	83.36	85.88	84.60	0.87	– 7.50
ESRGAN	83.53	82.76	85.28	83.99	0.86	– 8.10
Image augmentation	80.43	79.66	82.18	80.90	0.83	– 11.20
Dataset partitioning	78.93	78.16	80.68	79.40	0.81	– 12.70
VCNN (use CNN)	82.93	81.23	83.33	82.27	0.87	– 8.70
All preprocessing + augmentation	72.43	71.66	73.18	72.41	0.78	– 19.20

enhancement, and ESRGAN-based super-resolution), image augmentation, dataset partitioning, and the Volterra Convolutional Neural Network (VCNN) architecture. This study quantifies the impact of each phase on performance metrics (accuracy, precision, recall, *F1*-score, and AUC) [32], [33], [35] using the dataset of 12,000 RGB images ($224 \times 224 \times 3$) across 11 algal genera, with a 70:15:15 train-validation-test split and fivefold cross-validation. Additionally, the impact of directly feeding the raw dataset to the VCNN without preprocessing or augmentation is assessed. Results are summarized in Table 8, with the full framework achieving 91.63% accuracy, 90.86% precision, 93.38% recall, 92.10% *F1*-score, and an AUC of 0.94.

A. Ablation of preprocessing pipeline

The preprocessing pipeline comprises four sub-components: artifact removal, auto-deblurring, adaptive contrast enhancement (ACE-MSWDLHE), and ESRGAN-based super-resolution. Each was removed individually to assess its contribution.

- **Artifact Removal:** Bypassing artifact removal retains noise, increasing false positives for genera like *Oscillatoria* (5.6%) and *Nodularia* (6.1%) due to artifacts mimicking algal textures. Accuracy drops by 12.3–79.33%, and *F1*-score decreases to 79.33%, highlighting its role in ensuring clean images for feature extraction.
- **Auto-Deblurring:** Skipping auto-deblurring retains blurry images, obscuring morphological details like cell boundaries. This increases misclassifications (e.g., 7.2% false positives for *Anabaena* vs. *Nostoc*), reducing precision by 9.7–80.16% and accuracy to 81.93%. Deblurring is critical for restoring sharpness.
- **Adaptive Contrast Enhancement (ACE-MSWDLHE):** Omitting ACE reduces visibility of subtle features (e.g., pigmentation in *Karenia*), increasing false negatives by 6.8%. Recall drops by

7.5–85.88%, and accuracy falls to 84.13%, underscoring ACE's role in enhancing faint structures under varying illumination.

- **ESRGAN-Based Super-Resolution:** Removing ESRGAN retains low-resolution images, losing fine details (e.g., *Microcystis* cell textures). False negatives increase by 5.9%, reducing *F1*-score by 8.1–83.99% and accuracy to 83.53%. ESRGAN is essential for reconstructing high-frequency details.

B. Ablation of image augmentation

Image augmentation doubles the dataset from 6000 to 12,000 images using transformations (rotation, translation, shear, zoom, brightness) to simulate real-world variability. Without augmentation, the VCNN overfits to the smaller dataset, increasing false positives for minority genera like *Nodularia* (9.1%). Accuracy drops by 11.2–80.43%, and AUC falls to 0.83, confirming augmentation's role in enhancing generalizability.

C. Ablation of dataset partitioning

Dataset partitioning uses a 70:15:15 split with fivefold cross-validation to ensure robust training and evaluation. Bypassing partitioning (using the entire dataset for training without validation or testing) risks overfitting, increasing false negatives (e.g., 8.3% for *Karenia*). Accuracy drops by 12.7–78.93%, and AUC falls to 0.81, highlighting the necessity of partitioning for reliable performance estimates and class balance.

D. Ablation of VCNN architecture

Replacing the VCNN with a standard CNN eliminates Volterra Kernel Layers (VKLs), limiting feature extraction to linear operations. This increases misclassifications for morphologically similar genera (e.g., 7.1% for *Anabaena* vs. *Nostoc*), reducing accuracy by 8.7–82.93% and *F1*-score to 82.27%. VKLs are critical for capturing nonlinear feature interactions.

E. Direct input of raw dataset

Feeding the raw dataset (6000 images) directly to the VCNN without preprocessing or augmentation results in noisy, blurry, low-contrast, and low-resolution inputs with limited diversity. This severely degrades feature extraction, leading to increased misclassifications (e.g., 12.4% false positives for *Anabaena* vs. *Nostoc*). Accuracy drops by 19.2–72.43%, and *F1*-score falls to 72.41%, underscoring the synergistic importance of preprocessing and augmentation for robust HAB detection. The ablation results confirm the contribution of each component, with accuracy dropping sharply when preprocessing or augmentation steps are excluded, as described in Table 8.

F. Discussion

The ablation study confirms the critical role of each component in achieving the framework's high performance (91.63% accuracy, 92.10% *F1*-score). The preprocessing pipeline addresses real-world imaging challenges: artifact removal mitigates noise (12.3% accuracy impact), auto-deblurring restores sharpness (9.7% precision impact), ACE enhances faint structures (7.5% recall impact), and ESRGAN improves low-resolution images (8.1% *F1*-score impact). Augmentation (11.2% accuracy impact) and partitioning (12.7% accuracy impact) ensure generalizability and reliable evaluation, while VKLs (8.7% accuracy impact) enable nonlinear feature extraction for complex morphologies. Direct input of raw data results in a 19.2% accuracy drops, emphasizing the synergistic effect of all components. These findings align with the confusion matrix analysis (Figs. 8, 9) and performance metrics (Figs. 10, 11, 12, 13 and 14), reinforcing the framework's robustness for scalable, real-time HAB monitoring.

5.6 Impact of preprocessing techniques

The Preprocessing techniques are key for deep learning in harmful algal bloom (HAB) detection. A multi-preprocessing framework is proposed. It includes artifact removal to reduce noise. Auto-deblurring sharpens blurry images. Adaptive contrast enhancement improves clarity. Super-resolution techniques enhance input image quality. These steps optimize feature extraction.

- i. **Artifact Removal:** Preprocessing techniques are key for deep learning in harmful algal bloom (HAB) detection. A multi-preprocessing framework is proposed. It includes artifact removal to reduce noise. Auto-deblurring sharpens blurry images. Adaptive contrast

enhancement improves clarity. Super-resolution techniques enhance input image quality. These steps optimize feature extraction.

- ii. **Auto-Deblurring:** The Preprocessing techniques are key for deep learning in harmful algal bloom (HAB) detection. A multi-preprocessing framework is proposed. It includes artifact removal to reduce noise. Auto-deblurring sharpens blurry images. Adaptive contrast enhancement improves clarity. Super-resolution techniques enhance input image quality. These steps optimize feature extraction.
- iii. **Adaptive Contrast Enhancement (ACE):** The Microscopic images can be enhanced. Multi-Scale Wavelet Decomposition pairs with Local Histogram Equalization. Adaptive Contrast Enhancement (ACE) amplifies faint structures [34]. It balances light variations across the dataset. This aids in differentiating algal species. Experimental results showed a 7.5% recall improvement. The enhancement captures distinct patterns. It works well in complex environments.
- iv. **Super-Resolution Using ESRGAN:** Low-resolution images cause issues. Intricate algal morphologies lose details. ESRGAN reconstructs high-frequency details. It improves classification results. Including ESRGAN in the preprocessing pipeline raised the *F1*-score by 8.1%. This confirms enhanced image clarity. It also boosts model robustness.

5.7 Limitations and error analysis

While the proposed framework is effective, it is important to recognize certain limitations and error sources. The preprocessing pipeline is sensitive to extreme variations in image quality, which presents a primary challenge. A significant improvement in image clarity is achieved through artifact removal and auto-deblurring. Some of the essential algal structures may be interpreted as noise by these sensors, which can cause them to unintentionally lose critical characteristics. Although adaptive contrast enhancement is effective in improving visibility, it can sometimes over-amplify certain regions, leading to misclassification when algae exhibit low-intensity variations. The dependence on high-quality labeled datasets is another limitation. However, ESRGAN-based super-resolution cannot fully restore lost information in highly degraded samples, even when it enhances low-resolution images. Additionally, VCNNs demonstrate superior performance over conventional CNNs and AlexNet's. Since nonlinear feature extraction is integrated, it has a higher computational complexity. Due to its resource-constrained nature, it may not be suitable for use in real-time applications. The model is also prone to bias due to variations in lighting, water turbidity, and seasonal

changes, which could impair its generalizability. The framework's robustness and reliability will be enhanced if domain adaptation techniques and explainable AI strategies are incorporated in future work.

5.8 Discussion

The proposed VCNN framework outperforms the standard CNN (82.93%) and AlexNet (83.58%) with a classification accuracy of 91.63%, an error rate of 8.37%, and an $F1$ -score of 92.10%. Its nonlinear feature extraction and robust preprocessing pipeline enable superior handling of noise, artifacts, and low-resolution images. Benchmarking against models like Inception V3, U-Net, and AMDNN confirms the VCNN's ability to capture complex nonlinear interactions, achieving higher recall (93.38%) and $F1$ -score compared to U-Net's seasonal limitations or AMDNN's focus on transfer learning. The preprocessing pipeline integrates artifact removal. It includes auto-deblurring. It also uses Adaptive Contrast Enhancement (ACE) and ESRGAN. This ensures robust performance across diverse imaging modalities. The framework is scalable for real-time monitoring of harmful algal blooms (HABs). Benchmarking compared it to other approaches. These included Inception V3, U-Net, AMDNN, and traditional CNN-based models. The Volterra Convolutional Neural Network (VCNN) achieved an accuracy of 91.63%. It captures complex nonlinear interactions in algal image datasets. Traditional CNNs managed 82–84% accuracy. The AMDNN focuses on transfer learning. It has limited explainability compared to the VCNN. The VCNN handles noise artifacts effectively. It also adapts to diverse environmental conditions. It shows superior generalizability.

Furthermore, U-Net-based models, such as the dual U-Net proposed by Zhu et al., are limited in their ability to adapt to changing seasons. The proposed work, however, integrates adaptive contrast enhancement and ESRGAN super-resolution for robust performance across various imaging modalities and resolutions. Furthermore, generative techniques like StyleGAN2-ADA are primarily used to augment datasets, rather than performing classification directly. In addition to artifact removal, auto-deblurring, and feature-rich classification, the VCNN offers several other advantages. In comparison to existing methods, this comprehensive approach achieves a higher recall (93.38%) and $F1$ -score (92.10%), demonstrating its superiority in addressing critical challenges such as mixed species identification, low-resolution samples, and real-time scalability.

Table 9 shows that, unlike standard CNNs, which extract only linear features, VCNNs use Volterra Kernel Layers to capture nonlinear interactions. This enables the model to identify subtle structural patterns, thereby enhancing accuracy and robustness in classifying visually similar algal genera, even in noisy or low-resolution images.

Table 9 Differences between standard convolutional layers in CNNs and Volterra Kernel Layers (VKLs) in the proposed VCNN

Aspect	Standard CNN	Volterra Kernel Layers (VKLs)
Linear/nonlinear	Strictly linear convolution	Capable of nonlinear interactions when higher-order kernels are added
Filter representation	Standard convolution filters	Volterra kernels represent expansions (linear + nonlinear)
Flexibility	Restricted to linear approximations	More adaptable to complex, nonlinear patterns in the data

5.9 Generalizability and sensitivity analysis

The VCNN framework for harmful algal bloom classification was rigorously tested for generalizability under simulated environmental variations, including turbidity (10–30 NTU), illumination (80–120% brightness), and Gaussian noise ($\sigma = 0.05$). It achieved a robust accuracy of 89.45% under combined perturbations, significantly outperforming standard CNN (80.12%) and AlexNet (81.67%). However, challenges emerged with specific genera: *Karenia*'s accuracy dropped by 7.8% in high-turbidity (30 NTU) due to obscured morphological features, and *Microcystis* recall fell by 5.4% under low illumination, where pigmentation differences were less distinct. Misclassifications between *Anabaena* and *Nostoc* rose by 6.3% due to their similar filamentous structures. The preprocessing pipeline improved the $F1$ -score by 9.2% under noise conditions but struggled with extreme illumination, resulting in a 4.7% increase in false positives for *Oscillatoria* due to ACE overamplification. Future enhancements will explore domain adaptation and multi-modal data fusion to improve robustness across diverse ecological and imaging conditions.

The confusion matrix analysis further revealed that misclassifications increased under high-turbidity conditions, particularly between *Anabaena* and *Nostoc*, with a 6.3% increase in false positives for *Anabaena* misclassified as *Nostoc* due to their similar filamentous structures becoming less distinct in murky water. Noise sensitivity was effectively mitigated by the preprocessing pipeline, particularly through artifact removal and ESRGAN-based super-resolution, which reduced the impact of Gaussian noise by 9.2% in $F1$ -score compared to scenarios without preprocessing. However, extreme illumination variations (e.g.,

120% brightness) occasionally led to overamplification of low-contrast regions by the adaptive contrast enhancement module, resulting in a 4.7% increase in false positives for *Oscillatoria*.

These findings highlight the VCNN's strengths in handling moderate environmental variations, owing to its nonlinear feature extraction and robust preprocessing pipeline, while also underscoring its limitations in extreme conditions. The framework's reliance on high-quality morphological features makes it sensitive to severe turbidity or changes in illumination, which can obscure critical details. Seasonal variations, such as nutrient-driven changes in algal morphology, were not fully captured in the dataset, which was primarily sourced from freshwater ecosystems in Ethiopia. This limits generalizability to marine environments or regions with distinct ecological profiles, such as coastal waters affected by upwelling. Future work will address challenges with two key strategies. First, domain adaptation techniques will be used. These include adversarial training or transfer learning. They will fine-tune the VCNN for diverse geographic and seasonal conditions. Datasets from varied ecosystems will be leveraged. These include marine and brackish environments. Second, multi-modal data fusion will be implemented. This integrates image data with environmental variables. Variables include nutrient concentrations, temperature, and pH. These provide contextual cues for classification. The approaches will improve adaptability to real-world variability. They ensure reliable performance across regions, seasons, and imaging conditions. This strengthens the framework's utility for global HAB monitoring and management.

6 Conclusion and future directions

This work introduces a novel AI-driven framework for detecting and classifying harmful algal blooms (HABs), leveraging a Volterra Convolutional Neural Network (VCNN) integrated with an advanced preprocessing pipeline that includes artifact removal, auto-deblurring, adaptive contrast enhancement, and ESRGAN-based super-resolution. The framework addresses key challenges in HAB monitoring, such as noise, low-resolution images, and complex algal morphologies, achieving a classification accuracy of 91.63%, a precision of 90.86%, a recall of 93.38%, and an *F1*-score of 92.10% on a dataset of 12,000 RGB images across 11 algal genera. Compared to standard CNN (82.93% accuracy) and AlexNet (83.58% accuracy), the VCNN's Volterra Kernel Layers enable superior nonlinear feature extraction, reducing false negatives (e.g., 2.7% for *Karenia* vs. 8.9% for AlexNet) and false positives (e.g., 3.2% for *Microcystis* vs. 7.3% for CNN), as detailed in the confusion matrix analysis. This robustness, coupled with preprocessing techniques that enhance image clarity (e.g., 8.1% *F1*-score improvement via

ESRGAN), makes the framework a scalable, real-time solution for ecological monitoring and public health protection.

Despite its strengths, the framework has limitations. The preprocessing pipeline, although effective, is sensitive to extreme image degradation, where critical algal structures may be misinterpreted as noise, potentially leading to a loss of morphological details. For instance, the confusion matrix revealed misclassifications between *Anabaena* and *Nostoc* due to their similar filamentous structures, particularly in low-contrast images. The VCNN's higher computational complexity, driven by nonlinear feature extraction, may also limit its deployment in resource-constrained environments. Additionally, reliance on high-quality labeled datasets restricts generalizability, as ESRGAN cannot fully recover details in severely degraded samples, and seasonal or environmental variations may introduce biases.

Future work will address these challenges through several concrete directions. First, expanding the dataset to include a broader spectrum of algal species, encompassing both harmful and beneficial genera, will enhance our understanding of ecosystems and support comprehensive environmental management. Second, integrating explainable AI (XAI) techniques, such as attention heatmaps or feature importance analysis, will improve model transparency, fostering trust among policymakers and resource managers by clarifying decision-making processes. Third, deploying the framework on edge computing platforms, combined with the real-time integration of environmental variables (e.g., nutrient levels, temperature), will enable large-scale, in-situ monitoring of HABs. Finally, optimizing the VCNN architecture for lower computational complexity, potentially through kernel pruning or lightweight Volterra approximations, will enhance its suitability for real-time applications. These advancements will strengthen the framework's robustness, scalability, and impact on ecological conservation and public health, paving the way for more resilient water quality monitoring systems.

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Declarations

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